Case Adaptive Detection Models for Asbestos-Containing Building Materials Based on Hyperspectral Image Processing

Hyeji Sim, Jaehyung Yu, Lei Wang, Chanhyeok Park, Huy Hoa Huynh, and Hyun-Cheol Kim, *Member, IEEE*

***Abstract*—This study introduces case adaptive detection models for detecting asbestos-containing building materials employing short-wave infrared (SWIR) hyperspectral images. The models are optimized for indoor and outdoor conditions, which could be applied to asbestos detection in interior and exterior building materials. Samples were collected from the most probable types and forms of building materials for both asbestos-containing materials (ACMs) and non-ACMs to represent real-world conditions, utilizing the largest and most diverse dataset to date (11 types, four directions, and four forms). X-ray diffraction (XRD), scanning electron microscopy (SEM), and energy dispersive spectroscopy (EDS) analyses demonstrate that chrysotile was the main asbestos mineral in most of the ACMs, as evidenced by characteristic chrysotile diffraction peaks, fibrous serpentine morphology, and coincident MgSi elemental distributions. The spectral characteristics of building materials are manifested by mineral components for ACMs and non-ACMs, where asbestos absorption (MgOH) identifies ACMs. The indoor RF model identified critical absorption features at 22992350, 19231967, and 23662426 nm, corresponding to MgOH, OH, and CO_3^{2-} absorptions, achieving 95.3% accuracy. The outdoor RF model incorporated additional FeOH and AlOH bands, achieving 96.5% accuracy. Real-world validation at three asbestos abatement sites and one large-scale site demonstrated practical applicability, including scalability and computational efficiency with overall accuracies (OAs) of 92.1% for indoor condition and 92.5% for outdoor condition. Comparative analysis with existing benchmark methodologies confirmed superior performance, with our models significantly outperforming previous approaches. The models successfully detected ACMs across various material conditions, scales, and effectively verified complete removal postabatement with high computational speed. This approach delivers immediate practical value for compliance verification and quality assurance in real-world abatement scenarios and large-scale surveys, providing objective documentation for regulatory compliance while substantially reducing costs compared with conventional destructive sampling methods.**

***Index Terms*—Asbestos-containing materials (ACMs), chrysotile, hyperspectral image.**

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Hyeji Sim and Huy Hoa Huynh are with the Department of Earth, Environmental and Space Sciences, Chungnam National University, Daejeon 34134, South Korea (e-mail: hyeppy0519@naver.com; hoahcm198@o.cnu.ac.kr).

Jaehyung Yu is with the Department of Earth, Environmental and Space Sciences and the Department of Geological Sciences, Chungnam National University, Daejeon 34134, South Korea (e-mail: jaeyu@cnu.ac.kr).

Lei Wang is with the Department of Geography and Anthropology, Louisiana State University, Baton Rouge, LA 70803 USA (e-mail: leiwang@lsu.edu).

Chanhyeok Park is with the Department of Astronomy, Space Science and Geology, Chungnam National University, Daejeon 34134, South Korea (e-mail: pch304@naver.com).

Hyun-Cheol Kim is with the Center of Remote Sensing and GIS, Korea Polar Research Institute, Incheon 21990, South Korea (e-mail: kimhc@kopri.re.kr).

Data is available on-line at https://github.com/sayyou0519/ACM-detection-model.

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I. Introduction

ASBESTOS is a naturally occurring fibrous mineral classified into two distinct groups: serpentine and amphibole. The serpentine group is characterized by chrysotile, while the amphibole group comprises crocidolite, amosite, tremolite, actinolite, and anthophyllite [1]. Within the serpentine family, lizardite and antigorite exist as polymorphs exhibiting dense or fine-grained structures, contrasting with chrysotile's fibrous morphology that defines it as asbestos [2]. It is noteworthy that certain nonasbestos minerals, such as ferrierite, a member of the zeolite group, also exhibit fibrous structures that could potentially be misidentified as asbestos [3].

The extensive use of asbestos in building materials since the 1990s was primarily attributed to its exceptional insulation and flexibility properties [4], [5], [6]. However, the identification of carcinogenic properties in asbestos in the early 21st century led the International Agency for Research on Cancer to classify it as a Group 1 carcinogen [7]. This classification resulted in a worldwide ban on asbestos usage [8]. Nevertheless, the continued presence of asbestos-containing materials (ACMs) in existing structures necessitates ongoing detection and monitoring efforts.

Conventional methods for asbestos detection predominantly rely on destructive sampling techniques, followed by sample preparation and comprehensive physical–chemical analyses. These analytical approaches encompass X-ray diffraction (XRD), thermal analysis, polarized light microscopy (PLM), scanning electron microscopy (SEM), transmission electron microscopy (TEM), and X-ray synchrotron techniques [9], [10], [11], [12]. XRD and PLM methodologies are specifically employed to characterize asbestos minerals through the application of X-ray photons or by examining samples under specific illumination angles [10], [11], [12], [13].

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For detailed morphological and elemental composition analysis, TEM and SEM–energy dispersive spectroscopy (EDS) techniques utilize electron beam interactions with the sample [10], [11], [13]. Thermal analysis techniques evaluate structural and phase transformations by subjecting ACMs to elevated temperatures [11], while X-ray synchrotron microtomography enables non-destructive, high-resolution 3-D visualization of asbestos mineral morphology [14]. Although these traditional methodologies provide highly accurate and precise asbestos detection capabilities, their implementation is constrained by labor-intensive procedures, significant cost requirements, and limited spatial coverage for 2-D scanning or bulk sample analysis. Consequently, remote sensing and hyperspectral imaging (HSI) techniques have emerged as alternative approaches for large-scale asbestos detection and characterization.

The remote sensing technology offers significant advantages for ACM detection through its synoptic viewing capabilities, particularly in identifying asbestos-cement roof (ACR) using both uncrewed aerial vehicles (UAVs) and satellite platforms [15], [16], [17], [18], [19]. UAV-based applications utilizing high-resolution multispectral imagery have demonstrated successful ACR detection based on characteristic features including texture, structure, slope, and intensity patterns, achieving detection accuracies between 88% and 93% through machine learning implementations [18], [20], [21], [22]. Similarly, the hyperspectral data analysis has yielded approximately 90% accuracy in ACR detection across multiple studies [19], [23]. However, the limitation of nadir viewing in these remote sensing approaches restricts their application to roof materials exclusively. To address this constraint, ground-based close-range hyperspectral methods have been developed for ACM detection in both indoor and outdoor environments. Bonifazi et al. [24], [25] integrated the hyperspectral imaging analysis with micro-XRF for microscale mineral identification in construction materials (CMs). In a subsequent study, Bonifazi et al. [26] successfully differentiated between ACMs and non-ACMs—including fiberglass, wood, mortars, aggregate, rope, brick, and paper—utilizing principal component analysis (PCA) on short-wave infrared (SWIR) hyperspectral imagery, with validation through micro-XRF and microscopy techniques. Recent developments by Marzini et al. [27] introduced a portable asbestos mapping system based on near-infrared hyperspectral imaging (NIR-HSI) incorporating supercontinuum laser technology, though its application remains limited to six types of ACM and naturally occurring asbestos (NOAs).

Despite advances in ACM detection methodologies, substantial challenges persist in developing reliable detection systems for complex real-world applications. A critical limitation in current approaches is the requirement for comprehensive and diverse datasets that incorporate both positive and negative training samples to optimize model performance and reliability. Previous research efforts in CM asbestos detection have demonstrated notable constraints in their scope and applicability. UAV-based investigations have primarily concentrated on ACR detection in outdoor environments, often lacking sufficient diversity in training data collections [18], [20], [21]. Similarly, laboratory-based analytical approaches have been hindered by limited sample sizes and insufficient variation in training datasets.

A significant consideration in real-world applications is the diverse occurrence of ACMs, which extends beyond building interiors and exteriors to include waste materials. The presence of ACMs in construction waste presents heightened risk factors, particularly when materials are in degraded or powdered forms, as these conditions facilitate increased dispersion of asbestos fibers. The detection of asbestos components in construction waste materials thus represents a crucial parameter for urban regeneration projects. However, this critical aspect has been notably absent from previous investigations into ACM detection model development, representing a significant gap in current research methodologies.

This study aims to develop robust detection models for ACMs with broad applicability to real-world scenarios by incorporating diverse CMs in various forms and analyzing them across multiple viewing angles using hyperspectral imaging techniques (see Fig. 1). The experimental design encompasses three distinct types of ACMs and eight categories of non-ACMs, examined across multiple material states including fractured, grinded, stacked, and powdered forms. The data acquisition was conducted under both controlled laboratory conditions and field environments, with measurements taken at four distinct viewing directions to simulate various interior and exterior detection scenarios under different illumination conditions. The methodology incorporates comprehensive validation through multiple analytical techniques, including XRD, SEM, and EDS analyses.

II. Materials and Methods

A. Sample Selection and Preparation

As described, ACMs had been used worldwide for building materials such as textile and slate roof material containing asbestos from the 1970s to 1990s [28], [29], [30]. During the period, ACMs contained asbestos minerals from 10% to 15% as high as 20% in Europe [28]. ACMs in South Korea contain asbestos minerals 8%–14% for slate roof, 3%–6% for textile, 10% for interior wall panels (bamlight), and 8%–14% for asbestos compression exterior wall material.

This study selected asbestos textiles, ACRs, and bamlights as representative samples of ACMs to develop detection models of ACMs based on the hyperspectral imaging analysis. Moreover, the most common non-ACMs, such as gypsum textiles, bricks, gypsum boards, polystyrene insulations, tiles, asphalt roofs, and wood panels, were utilized as control groups that could be mixed with asbestos materials for both indoor and outdoor conditions (see Fig. 2 and Table I). Notably, the ACMs are more harmful in broken forms, such as fragmented and powdered in indoor conditions [31]. To simulate the real-world occurrence of CMs in harmful indoor conditions, we processed the samples in four forms for indoor ACM detection models: intact state, fragmented, grinded, and powdered forms (see Table I).

ACM samples were obtained from Daeji Environmental Industry Company Ltd., a certified asbestos inspection and

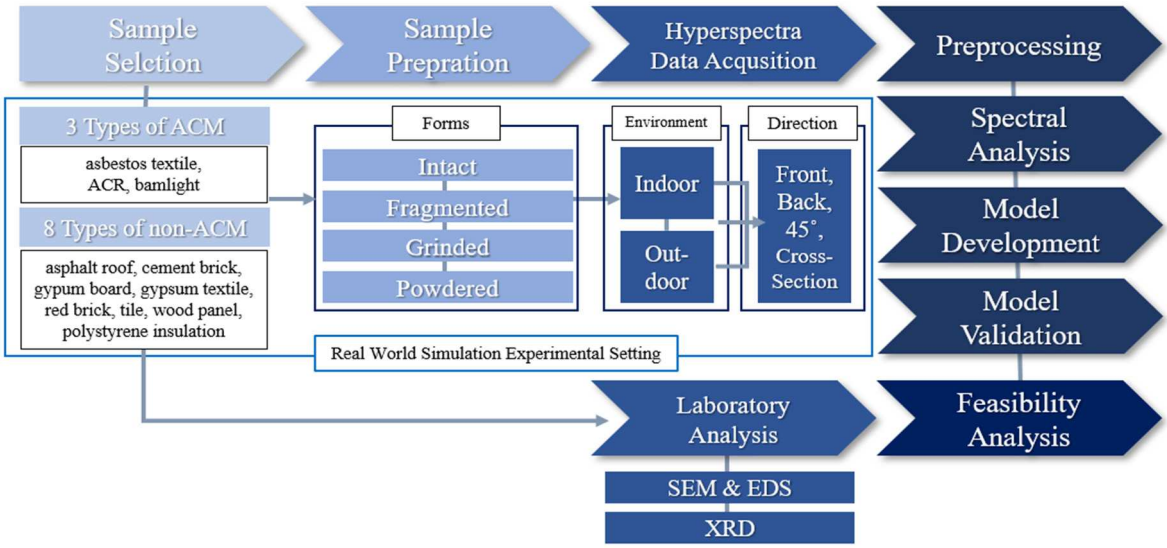


Fig. 1. Schematic flow of methodology.

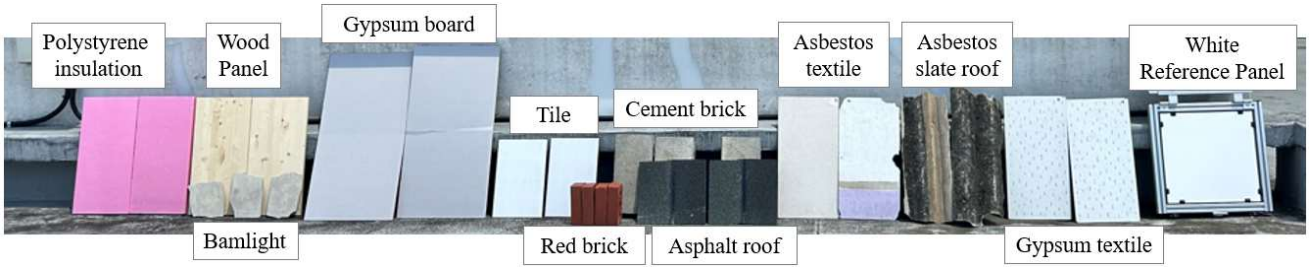


Fig. 2. Description of samples utilized for this study.

TABLE I
CONFIGURATION OF ACM AND NON-ACM SAMPLES USED IN THIS STUDY

Form	ACM	non-ACM*
Intact	asbestos textile, bamlight, ACR,	asphalt roof, cement brick, gypsum board, gypsum textile, red brick, tile, wood panel, polystyrene insulation
Fragmented	asbestos textile, bamlight, ACR	cement brick, gypsum board gypsum textile, red brick, tile
Grinded	asbestos textile, bamlight, ACR	cement brick, gypsum board, gypsum textile, red brick, tile
Powdered	asbestos textile, bamlight, ACR	-

*non-destructible non-ACMs including asphalt roof, wood panel, polystyrene insulation are excluded for sample processing.

removal company. These samples were collected during authorized asbestos removal operations at various sites, primarily government buildings and educational facilities constructed during the 1980s. An additional ACR sample was procured from a slate roof of an old residential structure in Jeongeup, Jeollabuk-do, South Korea. Non-ACM control samples were purchased from Rio the Warehouse, a building materials distributor located in Daejeon, South Korea.

B. SEM and EDS Analysis

The SEM analysis was conducted for the representative ACM and non-ACM samples as a validation method for the spectral analysis and for observing the microstructure of ACM and non-ACMs. Moreover, the EDS analysis was used to identify the specific chemical composition of corresponding microstructures in ACM and non-ACM samples. After identifying fibrous materials with a length of 5 μm or more and an aspect ratio (length-to-width ratio) of 3:1 or higher, magnesium (Mg) presence was verified within these materials. Subsequently, these materials were defined as asbestos.

This study employed SU7000, HITACHU for SEM analysis, and Ultim Max65, Oxford for EDS analysis (see Table II). The specifications of the instruments are summarized in Table II.

SEM is a type of optical microscope with a maximum resolution as high as 0.8 nm. This instrument enables magnification of the sample area by over 500 000 times, making it suitable for observing fine surface structures and morphology. Furthermore, the EDS analysis enables real-time confirmation of the composition and distribution of elements contained in the sample. A total of 11 different types of samples were used for SEM and EDS analyses.

Each sample was in powder or finely crushed form, thoroughly dried, coated with platinum for one minute, and then analyzed. This study observed samples under an

TABLE II
SPECIFICATIONS OF THE EQUIPMENT USED IN SEM AND EDS

Model Name	SU7000
Manufacturer	HIACHI
Resolution	0.8 nm @15kV, 0.9nm @1kV
Accelerating Voltage Range	0.1 kV ~ 30 kV
Probe Current Range	Max 200nA
Magnification	X 20 ~ X 2,000,000
Electron Gun	Hight brightness Schottky emitter
Model Name	Ultim Max65
Manufacturer	OXFORD
Resolution	127 eV at Mn Ka (130,000cps)
Active Area	65
Detector	Si Drift Detector (SDD)
Element of Analysis	Be (Atomic number: 4) ~ Cm (Atomic number: 96)

TABLE III
SPECIFICATIONS OF THE HYPERSPECTRAL CAMERA

Model Name	Spectral Camera SWIR	
Manufacturer	SPECIM	
Spectral Range	953-2562 nm	
Spectral resolution	FWHM: 12 nm	
Spectral bands	288	
Spectral Sampling	5.6 nm	
Spatial pixels	384	
Pixel size	24 * 24 µm	
Slith width	30 µm	
FOV	23°	
Size	Sensor	Power supply & Control Unit
	470*176*178 nm	300*190*130 nm
Weight	14 kg	5 kg

accelerating voltage of 10.00 kV and a working distance of 6.2 mm.

C. XRD Analysis

In addition to confirming the existence of asbestos in ACMs based on SEM analysis, the XRD analysis was additionally conducted to confirm ACMs and to figure out specific mineral components of asbestos. The confirmed ACMs are powdered using an agate mortar and pestle for sample preparation. Subsequently, the XRD analysis was conducted using a Rigaku Ultima IV X-ray diffractometer with Cu–Kα radiation (λ = 1.5406Å). The X-ray tube was operated at a voltage of 40 kV and a current of 30 mA. The diffraction angle (2θ) was set in the range of 3°–90° with a 0.02° interval, and the scanning speed was 10°/min. The data obtained using this instrument were analyzed using the Crystallographica Search Match program.

D. Hyperspectral Image Acquisition and Preprocessing

To build up detection models applicable for real-world indoor and outdoor cases, this study acquired hyperspectral images of ACM and non-ACM samples with various forms in indoor and outdoor at four different directions. The Specim SWIR hyperspectral camera was used to acquire the hyperspectral data. The camera acquires hyperspectral data with a spectral range from NIR to SWIR (953 ~ 2562 nm) with 288 bands at 12-nm spectral resolution and 6 nm spectral sampling (see Table III).

The outdoor hyperspectral image acquisition took place on June 23, 2023, from 11:30 A.M. to 1:00 P.M., utilizing sunlight as the light source. The camera-to-sample distance was set to approximately 5.3 m, and three scans were performed using an off-nadir approach. A white reference panel was positioned within the scenes for radiometric calibration. The frame rate is 40 Hz, and both spectral and spatial binning are set to 1. Exposure time and camera focus were carefully adjusted to prevent over-saturation. The direction of data acquisition was controlled by sample positioning, including vertical, 45° tilting, and stacked. The samples with different surfaces between front and back were additionally scanned for both sides (see Table IV).

TABLE IV
ACQUISITION CONDITIONS OF ACM AND NON-ACM MATERIALS WITH FOUR DIFFERENT ACQUISITION DIRECTIONS FOR INDOOR AND OUTDOOR HYPERSPECTRAL DATA

	Outdoor		Indoor	
Direction	ACM	non-ACM	ACM	non-ACM
Front	asbestos textile, ACR, bamlight	asphalt roof, cement brick, gypsum board, gypsum textile, red brick, tile, wood panel, polystyrene insulation	asbestos textile, ACR, bamlight	asphalt roof, cement brick, gypsum board, gypsum textile, redbrick, tile, wood panel, polystyrene insulation
Back	asbestos textile, ACR, bamlight	asphalt roof, cement brick, gypsum board, gypsum textile, tile, red brick, wood panel, polystyrene insulation	asbestos textile, ACR, bamlight	asphalt roof, cement brick, gypsum board, gypsum textile, redbrick, tile, wood panel, polystyrene insulation
45°	asbestos textile, bamlight, ACR, bamlight	asphalt roof, cement brick, gypsum board, gypsum textile, red brick, tile, wood panel, polystyrene insulation	asbestos textile, ACR, bamlight	asphalt roof, cement brick, gypsum board, gypsum textile, red brick, tile, wood panel, polystyrene insulation
Cross Section (stacked)	asbestos textile, ACR, bamlight	asphalt roof, cement brick, gypsum board, gypsum textile, tile red brick, wood panel, polystyrene insulation	asbestos textile, ACR, bamlight	asphalt roof, cement brick, gypsum board, gypsum textile, red brick, tile, wood panel, polystyrene insulation

The indoor hyperspectral image acquisition was performed under laboratory conditions using a horizontal line scanner mode at a distance of 0.6 m. The light source was two halogen lamps, and the reference panel was positioned within the scenes for radiometric calibration. The frame rate was 40 Hz, and the exposure time was 12.99. Both spectral binning and spatial binning were set to 1. The image capture process was performed using Lumo Scanner v.2.6 software (Specim Ltd., Finland). The same direction of data acquisition as outdoor data was used (see Table IV).

The acquired radiance images were converted to reflectance images by radiometric calibration using calibration panels within the scene. Moreover, the random noises were removed by the Tactical Hyperspectral Operations Resource (THOR) Spectral Smoothing tool and minimum noise fraction (MNF). The background pixels of the preprocessed images were further clipped, and the final preprocessed images were used for spectral analysis and detection model development. All hyperspectral image processing, including preprocessing and spectral analysis, were performed using ENVI Classic 5.3 (Harris Geospatial, Boulder, CO, USA).

E. Spectral Analysis

The preprocessed hyperspectral images were used for spectral analysis. The representative reflectance spectra for each type of CMs were derived based on the average spectra of the region of interest (ROI) pixels for each type of CMs. The representative reflectance spectra are further transformed to hull-quotient spectra to enhance absorption features manifested by mineral components. The spectral analysis was conducted for hull-quotient spectra of CMs, and a comparison between the representative spectra and the reference spectra of available CMs from United States Geological Survey (USGS) spectral library V7 and the Advanced Space-borne Thermal Emission and Reflectance Radiometer (ASTER) spectral library V 2.0 was carried out in the ENVI software.

To quantitatively evaluate spectral similarity between CMs and constituent minerals (calcite, chrysotile, and gypsum), the spectral correlation measure (SCM) and the spectral angle mapper (SAM) were calculated. SCM serves as a similarity index that captures spectral shape information, effectively measuring relative pattern similarity between reflectance spectra while partially compensating for brightness variations [32].

The SCM value γ is defined by the following equation:

$$\mathrm{SCM}(s_i, s_j) = \frac{\sum_{l=1}^{L}(s_{il} - \bar{s}_i)(s_{il} - \bar{s}_j)}{\sqrt{\sum_{l=1}^{L}(s_{il} - \bar{s}_i)^2 \cdot \sum_{l=1}^{L}(s_{jl} - \bar{s}_j)^2}} \quad (1)$$

where $r_{1,1}$ and $r_{2,1}$ represent the reflectance values at the lth wavelength for each spectrum and $\hat{r}_1$ and $\hat{r}_2$ denote the mean reflectance of each respective spectrum. L represents the total number of spectral bands employed in the analysis. For this study, 270 spectral bands spanning 1000–2500 nm were utilized for indoor measurements, while 91 bands covering 2000–2400 nm were applied for outdoor measurements. SCM values range from −1 to 1, with values approaching 1 indicating higher spectral similarity between compared curves.

Complementarily, SAM measures the angular difference between two spectral vectors, is largely invariant to overall brightness scaling and is relatively robust to local intensity fluctuations and noise. For two spectra s_i and s_j, SAM is defined as follows:

$$\theta = \arccos\left(\frac{s_i \cdot s_j}{\|s_i\|_2 \left\|s_j\right\|_2}\right)\left[^\circ\right]. \quad (2)$$

To ensure consistency with the SCM analysis, the same spectral band subsets were applied: 1000–2500 nm (270 bands) for indoor data and 2000–2400 nm (91 bands) for outdoor data. SAM values are expressed in degrees, ranging from 0° (identical spectra) to 90° (orthogonal spectra), where smaller angles indicate higher spectral similarity [33].

F. Data Selection for Model Development

The preprocessed reflectance hyperspectral images acquired in indoor and outdoor environments were used for training and validation data. The ACM data includes four different occurrence conditions, such as intact, fragmented, grinded, and powered, and the non-ACM data has three occurrence forms, including intact, fragmented, and grinded. Both ACM and non-ACM data have four different acquisition directions, including front, back, 45°, and cross section. Therefore, the hyperspectral data used in this study represent 16 different conditions for ACMs and 12 different conditions for non-ACMs to simulate the most possible cases in the real world for both indoor and outdoor conditions. Considering the illumination conditions and different usage, we developed independent detection models for indoor detection and outdoor detection. For both detection models, 28 combinations of occurrence and acquisition direction were categorized into two classes, ACM and non-ACM.

To build the ACM detection model, a total of 4400 pixels were randomly extracted and ACM and non-ACM classes from 28 different data combinations for model development and validation. Among these extracted pixels, 70% were utilized for model construction, and the remaining 30% served as validation data to assess the performance of the detection model.

G. Model Development and Validation

The ACM detection models were developed based on the random forest (RF) classification algorithm and binary logistic regression (BLR) model. RF is a machine learning classification technique that utilizes the ensemble of classification and regression trees (CARTs) to make predictions [34]. These trees are generated by randomly extracted subsets of training samples using a bagging approach [35]. In addition, RF provides out-of-bag (OOB) validation by utilizing data not used in the training set as test data, contributing to the accuracy of the RF classifier. RF as a classification technique has been widely adopted due to its notable advantages of high accuracy and processing speed [35]. Its main strength lies in the ability to rank variables based on their importance in decreasing the Gini impurity.

The BLR model is a statistical technique that predicts the probability of an event using a linear combination of independent variables. BLR has its merits in its simple form and easy interpretation of the coefficients, and with the stepwise selection of variables, it can find the best set of independent variables for the model.

1) Random Forest: This study constructed and validated the RF model to detect ACMs and non-ACMs in indoor and outdoor conditions. In addition, the RF method was utilized to select crucial spectral bands to detect ACM. The RF assessed the importance of variables through the Gini coefficient and mean decrease in accuracy. The Gini coefficient, also known as the impurity index, is employed in selecting the most important variables for classifying samples within the decision tree algorithm. Following the selection of major bands based on the Gini coefficient, the spectral variation of ACM within those wavelength ranges was analyzed.

To create an RF tree, the number of decision trees to be generated (N_{tree}) and the number of variables to be tested for the best split when growing the trees (M_{try}) must be specified. In this study, N_{tree} was set at 500 as previous studies figured out that the errors are stabilized at less than 500 trees for most of cases [36].

Model performance was evaluated using metrics derived from the confusion matrix, which consists of true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). Overall accuracy (OA), representing the proportion of correct classifications, was calculated as follows:

$$\text{Overall Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{3}$$

The kappa coefficient (k), which accounts for chance agreement, was determined by

$$k = \frac{P_0 - P_e}{1 - P_e} \tag{4}$$

where P_o is the observed agreement and P_e is the expected agreement by chance. The coefficient ranges from -1 to 1, with higher values indicating stronger agreement.

Additional metrics included the user's accuracy (proportion of TPs among predicted positives), producer's accuracy (proportion of actual positives correctly identified), commission error (FP rate), and omission error (proportion of missed positives). These metrics comprehensively assessed the RF model's performance in ACM classification.

Specificity measures the model's ability to accurately classify non-ACM materials and is expressed as follows:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \tag{5}$$

Precision (positive predictive value) indicates the proportion of correctly identified ACM materials among all materials classified as ACM and is calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{6}$$

These metrics are particularly effective in evaluating the robustness of models using hyperspectral data, where accurate differentiation between ACM and non-ACM materials is critical. In addition, the $F1$-score was employed to assess the balance between precision and recall, especially in the case of class imbalance

$$F1\text{-Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{7}$$

These combined metrics comprehensively assessed the RF model's performance in ACM detection under varying environmental conditions.

2) BLR Model With Stepwise Variable Selection: The ACM detection model using BLR was constructed using reflectance from hyperspectral images within the range of 1000–2500 nm (NIR-SWIR) as independent variables. BLR is composed of one or multiple independent variables between the independent and dependent variables [35]. The logistic regression analysis was employed based on the binary operator to predict the probability of matching spectral reflectance as follows:

$$P = \frac{e^{(C + x_1\beta_1 + x_2\beta_2 + \cdots + x_n\beta_n)}}{1 + e^{(C + x_1\beta_1 + x_2\beta_2 + \cdots + x_n\beta_n)}} \tag{8}$$

where P is the probability of the ACM class occurrence, C is the constant value of the equation, $\beta_1 \sim \beta_n$ are the coefficients of the predictor variables, and $x_1 \sim x_n$ are the reflectance values of the selected spectral bands. The range of P values lies between 0 and 1, where the pixels with P values equal to or greater than 0.5 are considered as ACM [37]. The best BLR model was validated based on the Hosmer and Lemeshow test, Wald statistics, and pseudo-R^2 values of "Nagelkerke" and "Cox and Snell." The Hosmer and Lemeshow test assesses the fitness of logistic regression by validating the log-likelihood ratio between the predicted and observed event occurrence rates within subgroups. Subsequently, the test statistic "H" is computed along with a p-value test to evaluate the predictive ability of the model. The pseudo-R^2 values of "Nagelkerke" and "Cox and Snell" were used to assess the validity of the constructed model [38]. In the following equation, $L(M_{\text{intercept}})$ represents the ratio of log-likelihood when the model has no explanatory variables, whereas $L(M_{\text{full}})$ represents the log-likelihood ratio when the model includes explanatory variables. These pseudo-R^2 values range from 0 to 1, where a value closer to 1 indicates a better effectiveness of the model

$$R^2 = 1 - \left\{\frac{L(M_{\text{intercept}})}{L(M_{\text{full}})}\right\}^{\frac{2}{N}}, \quad \text{Cox and Snell} \tag{9}$$

$$R^2 = \frac{1 - \left\{\frac{L(M_{\text{intercept}})}{L(M_{full})}\right\}^{\frac{2}{N}}}{1 - L(M_{\text{intercept}})^{\frac{2}{N}}}, \quad \text{Nagelkerke.} \tag{10}$$

In addition, the Wald statistic [(coefficient/standard error)^2] was utilized to assess the statistical significance and importance of each explanatory variable within the constructed model [38]. The classification performance of the BLR model was evaluated using the same metrics as the RF model, including the confusion matrix, OA, kappa coefficient, commission error, omission error, user's accuracy, and producer's accuracy. In addition, specificity (TN rate), and $F1$-score were also calculated to provide a more comprehensive evaluation of classification performance. The OA and kappa coefficient were calculated following the same approach as described in the RF model section. Through the previously mentioned validation methods, the most statistically significant and optimal equations for detecting ACM were selected for indoor and outdoor conditions, respectively. All statistical analyses, including the construction and validation of the logistic model, were performed using software packages including ENVI 5.6.2, IDL 8.5, IBM SPSS 26, and Python 3.9.7. Subsequently, the derived logistic regression model was applied to the hyperspectral images to map the distribution of ACMs at various conditions. Furthermore, the final results of imaging ACMs were evaluated using metrics such as confusion matrix, OA, kappa coefficient, and error rates.

H. Feasibility Analysis

To evaluate the detection model's real-world applicability, we conducted field validation at the Natural Science Building 3, Chungnam National University. This building, constructed in 1989 during the period when ACMs were commonly used in construction, underwent comprehensive asbestos abatement from December 2022 to January 2023. The project involved replacing all identified ACMs with non-ACMs across approximately 12 072 m^2, of which 5754 m^2 contained asbestos



Fig. 3. Images of asbestos removal operations at field validation buildings used for feasibility analysis.



Fig. 4. Real-world validation sites used for hyperspectral model performance assessment in indoor (a) and (b) and outdoor (c) environments.

materials (see Fig. 3) [39]. The preabatement and postabatement conditions provided an optimal scenario for real-world validation testing. Three validation sites were selected: two indoor locations consisting of a second-floor hallway ceiling (asbestos textile replaced with gypsum textile) and an ICP laboratory wall (bamlight replaced with gypsum board), and one outdoor location on the building roof (ACR replaced with corrugated iron roofing) (see Fig. 4). SWIR hyperspectral images were acquired at these locations in October 2022 (preabatement) and July 2025 (postabatement). This temporal comparative analysis not only demonstrates the model's field applicability but also validates its utility for verifying the completeness and effectiveness of asbestos abatement operations. The hyperspectral imaging data for validation sites were acquired using a Specim SWIR hyperspectral camera system. All acquired images underwent radiometric correction to convert digital numbers to reflectance values for subsequent spectral analysis. For indoor acquisitions, two 500-W halogen lamps provided consistent illumination conditions, with a Spectralon[1] white reference panel positioned within each scan area for reflectance calibration. Acquisition parameters were configured with a frame rate of 40 Hz and an exposure time of 22 ms to optimize signal-to-noise ratio while preventing spectral saturation. Outdoor hyperspectral imaging of ACR materials was conducted from the fourth-floor rooftop of the storage facility using natural sunlight. The imaging geometry maintained a standoff distance of approximately 5 m between the sensor and target materials. Five independent scans were acquired using an off-nadir viewing angle (approximately 15° from vertical) to minimize specular reflectance artifacts commonly encountered with smooth roofing surfaces. Acquisition parameters included a frame rate of 40 Hz with spectral and spatial binning factors of 1. ACM surveys require large-scale detection capabilities for practical outdoor applications where identification must be performed across extensive areas. To evaluate the scalability of the proposed method, we conducted a feasibility test over a populated town area in Jeonmyeon, Geumsan-gun, Chungcheongnam-do, South Korea. The hyperspectral data were acquired on October 1, 2025, through four overlapping scans covering the region. Standard preprocessing procedures, including radiometric correction and noise removal, were applied, followed by mosaicking to generate a large-scale scene with spatial dimensions of 7500 × 384 pixels. This scene encompasses numerous aged buildings constructed with various materials, including wood, cement tile, steel, shingle, and ACR, confirmed through official building registry records from the local government [40] (see Fig. 5). To evaluate detection effectiveness and computational efficiency across different scales, we performed comparative analysis on multiple test sites spanning various spatial extents and data volumes. The test sites range from small-scale targets of several meters at 492 × 384 pixels [ACR, Fig. 4(c)] to large-scale areas exceeding 100 m at 7500 × 384 pixels (town site, Fig. 5). Accuracy assessment across these sites evaluates the detection performance relative to spatial scale, while processing time measurements quantify computational efficiency. All experiments were conducted on a Windows 10 system equipped with an Intel Core i7-9700K CPU and 128-GB RAM. The processing speed was reported in terms of sec/img and resolution-invariant milliseconds per megapixels. Recognizing that real-world ACM applications typically involve complete structural elements (walls, roofs, and ceilings) rather than fragmented portions, we implemented a dual-level assessment framework. While pixel-based accuracy metrics provided foundational performance indicators, we supplemented these with object-level detection metrics to better represent practical field conditions. Using a majority-vote rule (≥ 50% pixel threshold), we calculated detection rate and false alarm rate to assess the proportion of correctly identified ACM objects and the proportion of non-ACM objects incorrectly classified as ACM, respectively. These object-based indicators offer operationally relevant performance measures that align with real-world ACM detection scenarios, ensuring consistency between spectral classification results and practical deployment outcomes.

III. Results

A. SEM and EDS and XRD Analyses

To determine the presence of asbestos in scanned building materials and to define the asbestos minerals contained therein, SEM and EDS analyses were utilized. In the case of ACMs, the identification of asbestos minerals contained in ACMs was carried out for powdered and intact ACMs based on morphology. For non-ACMs, intact ACMs were scanned. The SEM analysis was conducted at a magnification of 1000× and 400×. The SEM analysis revealed fibrous structures characteristic of chrysotile in all ACM samples (see Fig. 6). These fibers exhibit distinctive curly morphology with lengths exceeding 5 μm and width-to-length ratios greater than 3:1. Chrysotile's characteristic curled and coiled morphology, which distinguishes it from the straight, needle-like structure of amphibole asbestos minerals, is predominantly found in CMs [41]. This unique morphology derives from chrysotile's fundamental crystal structure within the serpentine group. The mineral structure comprises silicate tetrahedral layers bonded

[1]Registered trademark.

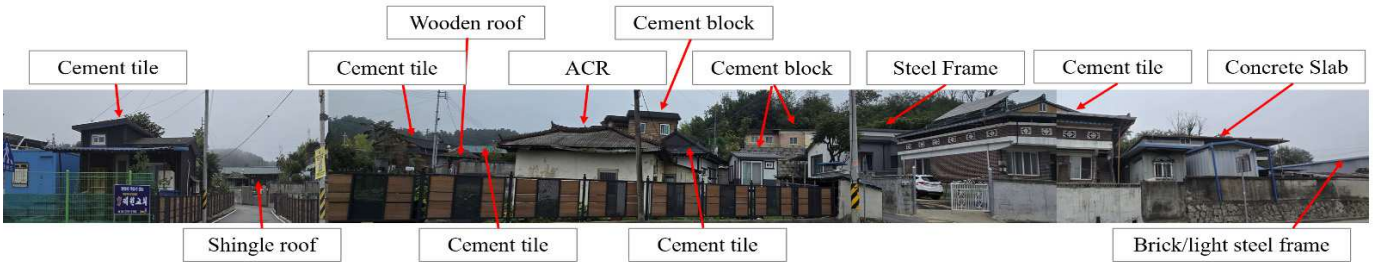


Fig. 5. RGB image of the large-scale validation site covering a town sector in Geumsan, South Korea, showing various building materials.

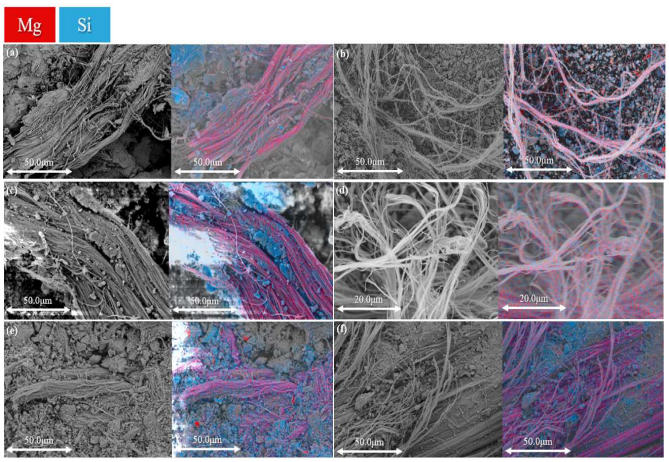


Fig. 6. SEM and EDS Mg/Si images of ACMs. (a) Nondestructed surface of ACR, (b) powdered ACR, (c) nondestructed surface of asbestos textile, (d) powdered asbestos textile, (e) nondestructed of bamlight, and (f) powdered bamlight.

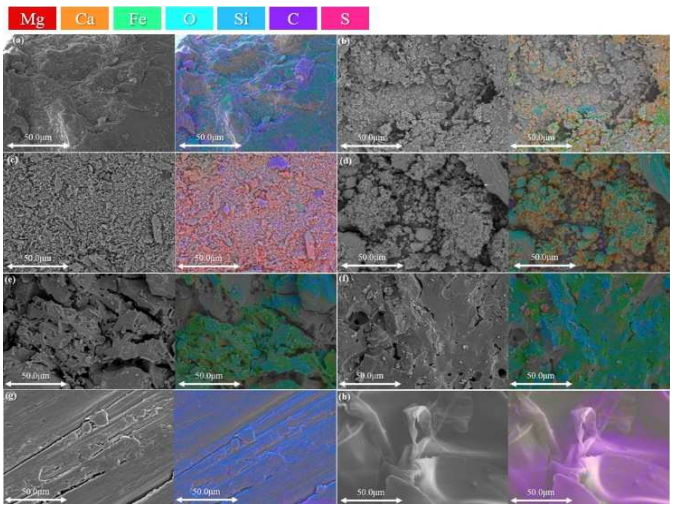


Fig. 7. SEM and EDS Mg/Si images of non-ACMs. (a) Asphalt roof, (b) cement brick, (c) gypsum board, (d) gypsum textile, (e) red brick, (f) tile, (g) wood panel, and (h) polystyrene insulation.

to brucite-like octahedral layers $[MgO_2(OH)_4]^{6-}$, forming a composite arrangement [2]. These layers can deform into undulating configurations or develop fibrous morphologies, held together by weak hydrogen bonding [42]. While this structural arrangement determines chrysotile's characteristic physical properties, other serpentine group minerals, such as antigorite and lizardite, maintain a columnar structure and a platy structure, respectively, with minimal fibrous characteristics, highlighting a distinct morphological difference from chrysotile [43]. Comparing SEM images of powdered and intact forms, the asbestos in powered ACMs was separated into thinner fibrous mineral bundles with a thickness ranging from 0.05 to 0.1 μm [see Fig. 6(b), (d), and (f)]. On the other hand, the asbestos in intact ACMs existed in bundles [see Fig. 6(a), (c), and (e)], with the thickness of the bundles being approximately 80–90 μm. Such characteristics suggest that the destruction of building materials would cause powered ACMs to break asbestos bundles into individual fibers in smaller sizes. However, fibers shorter than 5 μm also need to be detected because they are not free from carcinogenic risk [44]. The spatial distribution of asbestos minerals was characterized using EDS elemental mapping of Mg and Si. Chrysotile ($Mg_3Si_2O_5(OH)_4$), identified as the predominant asbestos mineral in these samples, contains Mg and Si as essential chemical components resulting from serpentinization of ultramafic rocks under hydrothermal alteration [45]. The coincident distribution of Mg and Si serves as a diagnostic chemical signature for chrysotile identification. EDS mapping thus provides complementary chemical validation to the morphological characteristics observed by SEM and the crystallographic data obtained from XRD analysis. None of the non-ACM showed a characteristic fibrous structure of asbestos minerals. In the case of the asphalt roof, a thin film-like structure was observed, with elements *C* and Fe predominantly present in that area. This can be explained by the one of the major components of asphalt roof is a graphene(C) [see Fig. 7(a)] [46]. The commonly used building materials such as cement bricks [see Fig. 7(b)], gypsum textiles [see Fig. 7(d)], and red bricks [see Fig. 7(e)] showed characteristic forms manifested by mineral component of Portland cement including hexagonal plates of portlandite, octahedral forms of periclase, plate crystals of tobermorite, and aggregate. It infers that these building materials are likely to have been produced in combination with the Portland cement, with elements Ca and Fe being the most prominently expressed. These elements are presumed to originate from the calcium oxide (CaO) and iron oxide (Fe_2O_3) [47]. In the case of gypsum board [see Fig. 7(c)], SEM images revealed distinct gypsum's monolithic crystal forms. The gypsum's appearance was confirmed by EDS analysis showing predominant distribution of Ca

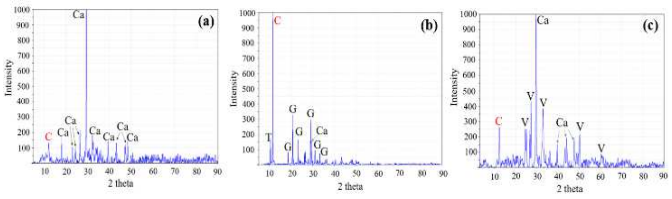


Fig. 8. XRD analysis results of ACMs (C-chrysotile, Ca-calcite, G-gypsum, T-tridymite, and V-vaterite). (a) Asbestos textile, (b) ACR, and (c) bamlight.

and S associated with the chemical composition of gypsum ($CaSO_4 \cdot 2H_2O$). In contrast, for gypsum textiles, it was difficult to confirm the crystalline form of gypsum unlike gypsum boards [see Fig. 7(d)]. In the SEM images of tile [see Fig. 7(f)], wood panel [see Fig. 7(g)], and polystyrene insulation [see Fig. 7(h)], no mineral crystals were observed. The surface of the wood panel displayed the structure of wood fibers, predominantly showing the presence of carbon due to organic materials [see Fig. 7(g)]. The polystyrene insulation [see Fig. 7(h)] also primarily showed carbon, resulting from its polymer base. In the case of the tile [see Fig. 7(f)], Si and Fe were predominantly detected, which seems to be due to SiO_2 and Fe_2O_3, the major components in the frit, the tile's primary material [48]. The XRD analysis was conducted to define the asbestos minerals present in the ACMs (see Fig. 8). The XRD analysis revealed a common peak of chrysotile at 10.5°–13.5° 2θ in ACMs. This result confirms the results from SEM and EDS analysis. In addition, peaks of gypsum, tridymite, and calcite were observed in asbestos textiles, while calcite and vaterite peaks were detected in bamlight. In ACR, peaks of calcite were noted. Gypsum, calcite, and vaterite are known minerals commonly used in building materials, suggesting their presence as mixtures during CM production [49]. Tridymite is also commonly found in glassy materials and is presumed to be present as an admixture in building material production [50].

B. Spectral Analysis

1) Indoor: ACMs' spectrum analysis results (see Fig. 9) showed that absorption characteristics were commonly observed at 1400, 1900, and 2327 nm for all ACMs. The 1400- and 1900-nm regions are associated with hydrolysis and water components. Notably, the absorption at 2327 nm is related to Mg–OH bonding manifested by asbestos minerals. Asbestos textiles exhibited the highest average reflectance among the ACMs and have characteristic doublet absorptions near 2300 nm. This absorption is associated with Al–OH which may be manifested by mineral components such as aluminosilicates during the manufacturing process. However, the doublet structure was not observed in the bamlight and ACR in this region. The bamlight has the deepest Mg–OH absorption among the ACMs, indicating the highest content of Mg–OH bonding. Comparing the ACMs spectra with the chrysotile library spectrum, the absorption features match each other well at hydroxyl and Mg–OH bonding, while the library spectra showed doublet Fe–OH absorption. Notably, the absorption depth was deepest at Mg–OH (2327 nm)

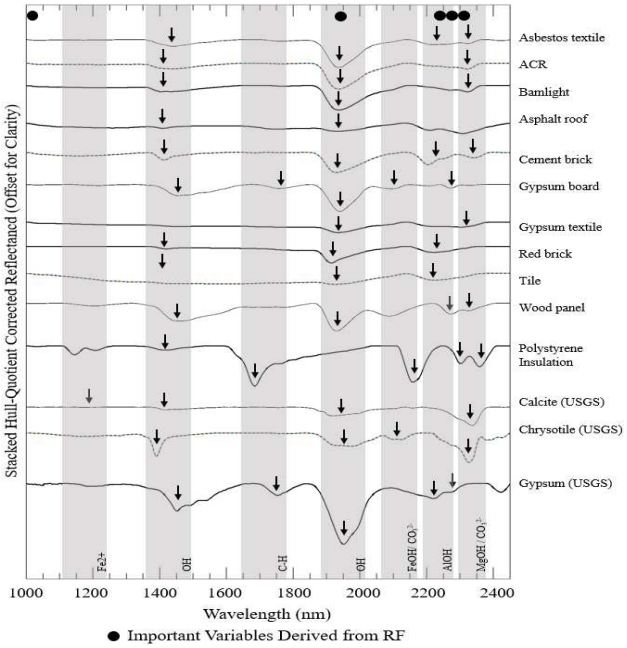


Fig. 9. Representative SWIR hull-quotient spectra (offset for clarity) of the ACMs and non-ACMs in indoor conditions.

among the three common absorption features for both ACMs and chrysotile. Given that the presence of chrysotile is confirmed by both SEM and XRD analysis, combining the three common absorptions would be useful to define the spectral characteristics of ACMs. Gypsum board and gypsum textile exhibited common absorptions at 1400 and 1900 nm (OH). Gypsum textile showed a shallow CO_3^{2-} absorption at 2300 nm. Gypsum board exhibited absorptions at 1700 nm (CH) and 2100 nm (Fe–OH), as well as a doublet absorption feature in the range of 2200–2350 nm (Al–OH–Mg–OH) (see Fig. 9). The library spectrum of gypsum can be defined by Fe^{2+}, C–H, Al–OH–Mg–OH, and CO_3^{2-} absorptions (see Fig. 9). Among these, AlOH–MgOH doublet absorption and CO_3^{2-} absorption are known to be presentative spectral characteristics of gypsum minerals, which overlap with gypsum CMs as confirmed by SEM analysis. Cement brick exhibited similar absorption characteristics with gypsum CMs except for the existence of CH absorption. Red brick only showed hydrolysis and Al–OH absorptions (see Fig. 9). Cement brick matched the spectral characteristics of calcite, the main source material for cement. On the other hand, the red brick only showed clay absorption because the red bricks are mainly made of clay and sand, and sand is mainly composed of quartz, which has no absorption features. Among the non-ACMs, spectra of asphalt roof, tile, wood panel, and polystyrene insulation were analyzed (see Fig. 9). In the case of asphalt roof, no absorption characteristics were observed, and the reflectance value was very low compared to other building materials. Tiles showed weak absorption characteristics at 2200 nm (Al–OH). Wood panels exhibited significantly deeper absorption in the OH region associated with moisture originating from organic materials [50]. Polystyrene insulation showed a doublet absorption at 1100–1200 nm (Fe^2) and a deep absorption at 1650 nm.

This is believed to be associated with petroleum's C–H stretch, as polystyrene foam is the main material used. Compared to non-ACM spectra, ACM commonly shows absorption features around 2300 nm (Mg–OH) with the lowest point typically at 2327 nm. Although most of the non-ACMs did not show the Mg–OH absorptions, gypsum board, gypsum textile, and cement brick exhibit similar absorption characteristics near 2300 nm. The notable difference in the Mg–OH absorption can be explained by the absorption depth, where ACMs generally show deeper and narrower absorptions, indicating the substantial abundance of Mg–OH components. The spectral characteristics of ACMs vary by form, although the general pattern is maintained. In general, the absorption gets deeper as the size of fragments decreases (see Fig. 10). Asbestos textile exhibited single form absorption characteristics when intact, but a doublet structure appeared in other forms. For ACR, unlike asbestos textile and bamlight, the doublet structure was not observed, but the absorption characteristics were deeper in the damaged state. For bamlight, all forms exhibited doublet structure absorption characteristics, with the deepest absorption in the grinded form. It infers that the detection model using intact form may not be able to find powdered or grinded asbestos in indoor conditions which have the most harmful effect to the human body system [32]. This study has figured out these characteristics for the first time. Furthermore, the spectral characteristics also change by acquisition direction [see Fig. 10(b)]. The doublet structure in the asbestos band of the asbestos textile varied across the four directions. The front view showed similar absorption depth for the doublet, while the back and cross-sectional view had a deeper-front doublet, and the 45° angle had a deeper back doublet structure. For ACR, a single absorption feature appeared in the asbestos band, with the deepest absorption in the front view. Bamlight showed a single absorption feature in the asbestos band, with the deepest absorption in the back view. In addition, the Fe^{2+} absorption feature was evident in the cross section. It indicates that the detection of asbestos in indoor environments has to take into consideration the acquisition direction.

2) Outdoor: The spectrum acquired in outdoor environments exhibited strong atmospheric absorptions near 1400 and 1900 nm. Therefore, the spectral characteristics for ACMs and non-ACMs in outdoor conditions were analyzed for spectral regions beyond 2000 nm, where the mineral absorption characteristics are well represented. Outdoor spectra showed significantly weaker spectral characteristics than those acquired indoors, although the spectral characteristics are similar to indoor conditions. ACM spectra exhibited Mg–OH absorption in all ACMs like those of the indoor environment, but much shallower. Moreover, CO_3^{2-} absorption was found for all cases (see Fig. 11). In the case of gypsum-containing CMs, both indoor and outdoor spectra were similar except for gypsum textiles, which exhibited Mg–OH and CO_3^{2-} absorptions (see Fig. 11). Cement brick showed the similar absorption as the indoor condition, while red brick spectra showed a longer wavelength-shifted absorption from 2200 to 2370 nm (see Fig. 11). Asphalt roof and tile showed patterns similar to the absorption characteristics observed indoors. The polystyrene insulation exhibited absorption characteristics of

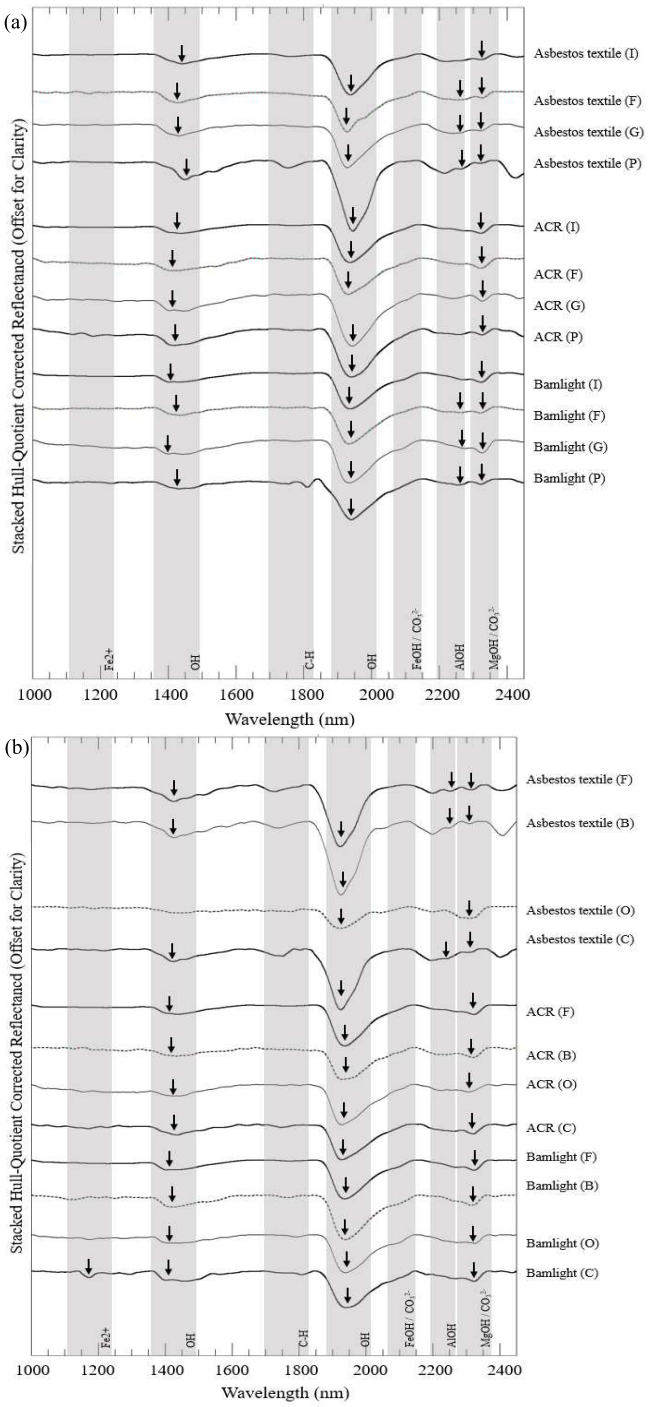


Fig. 10. (a) Hull-quotient spectra of ACMs in four different forms in indoor conditions. I: intact, F: fragmented, G: grinded, and P: powdered. (b) Hull-quotient spectra of ACMs in four different acquisition directions. F: front, B: back, O: 45° angle, and C: cross section (stacked).

CO_3^{2-} (2150, 2320 nm) both indoors and outdoors, but the doublet structure was not observed at 2370 nm in outdoor environments. Wood panels showed significant differences between indoor and outdoor spectra, displaying only CO_3^{2-} absorption characteristics for outdoors (see Fig. 11). In general, the spectral characteristics of asbestos textiles by various acquisition directions showed similar doublet absorptions at the Mg–OH band, except for the cross-sectional view, which showed a

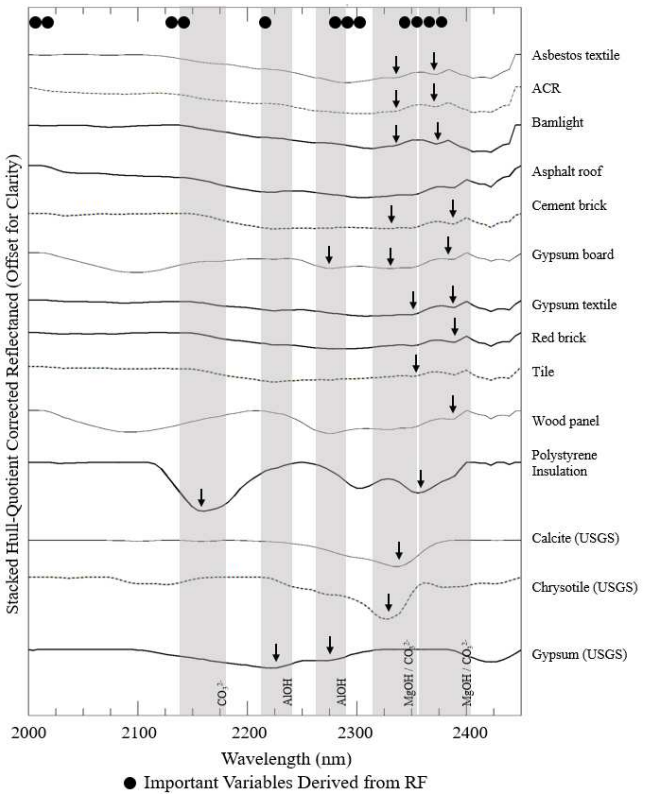


Fig. 11. Representative SWIR hull-quotient spectra (offset for clarity) of the ACMs and non-ACMs in outdoor conditions.

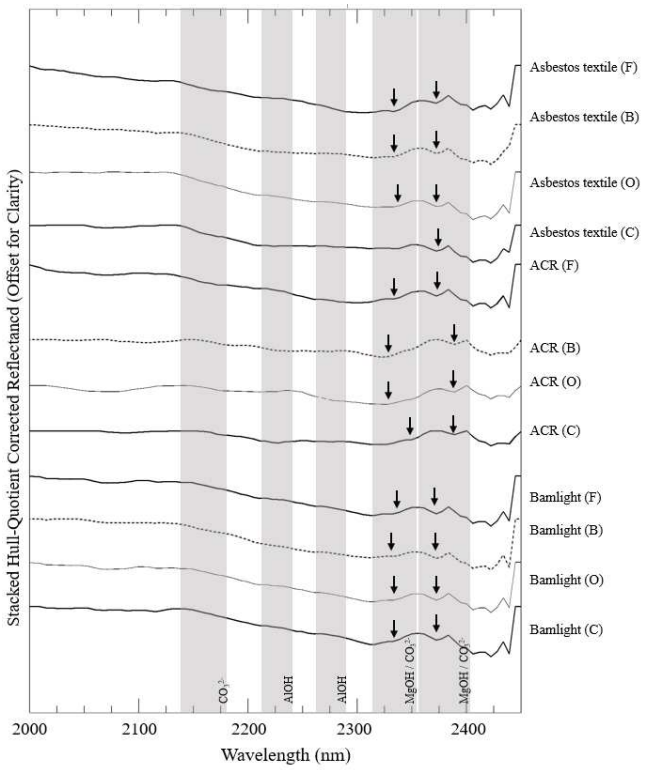


Fig. 12. Hull-quotient spectra of ACMs in four different acquisition angles in outdoor conditions. F: front, B: back, O: 45° angle, and C: cross section (stacked).

single absorption (see Fig. 12). In contrast, the spectra of ACR and bamlight showed single absorption characteristics for all directions except for the front view. When captured from the front, the spectra showed a doublet absorption characteristic in the Mg–OH band, differing from the spectra obtained at other directions. It confirms that various acquisition directions should be considered for the development of detection models. More importantly, the outdoor survey often faces wasted CMs where the positioning is not set to front view only. Therefore, the detection model using various acquisition directions would provide better detection efficiency.

3) Spectral Similarity: This study evaluated spectral similarity between CMs and reference minerals (chrysotile, gypsum, and calcite) from the USGS spectral library using SCM and SAM techniques. Similarity thresholds were established at SCM values greater than ± 0.8 and SAM values below 6° (0.1 rad). The results were visualized as wavelength-dependent distributions, enabling identification of mineral components in CMs through their characteristic absorption features. ACM samples under indoor conditions showed significant spectral similarity with chrysotile, particularly in the 2300–2350-nm region as demonstrated by both SCM and SAM distributions (see Figs. 13 and 14). This wavelength range corresponds to the characteristic Mg–OH bonding features observed in the reflectance spectra, indicating chrysotile's strong spectral influence on these materials. The SAM analysis revealed near-zero angular differences between ACM and chrysotile spectra in the range of 1000–1300 and 1500–1800 nm, further confirming their strong spectral resemblance. Asbestos textile samples displayed generally lower and more irregular SCM values, though they maintained moderate similarity levels (0.6–0.8) in the Mg–OH region, likely because their multicomponent composition diluted the distinctive spectral features of asbestos minerals [see Fig. 13(b)]. In addition, SCM values exceeding 0.6 were observed in the range of 2350–2400 nm, potentially indicating carbonate contributions. In this particular wavelength region, however, the SAM values occasionally showed higher similarity to calcite and gypsum than to chrysotile [see Fig. 14(a)–(c)]. Non-ACM samples showed lower spectral similarity with chrysotile than ACMs, with SCM values below 0.8 across most wavelengths [see Fig. 13(d)–(k)], though SAM values indicated significant overlap. Most non-ACMs displayed similarity in SAM values at chrysotile's 2300–2350-nm absorption band, except polystyrene insulation, red brick, and wood panel. Cement brick demonstrated high similarity with calcite, while gypsum board and gypsum textile showed strong correlations with reference gypsum spectra (SCM: 0.75–0.8 and SAM: 2°–5°) for CO_3^{2-} and water absorptions, confirming the presence of these mineral components [see Fig. 13(e)–(g)]. In outdoor, ACMs exhibited lower spectral correlation with chrysotile around 2350 nm compared with indoor conditions, which appears to be due to the weakened spectral characteristics, as previously discussed. In this spectral region, the presence of CO_3^{2-} absorption features also contributed to increased correlation with gypsum-based materials [see Figs. 15 and 16(a)–(c)]. Nevertheless, most ACMs maintained SCM values above 0.6 with chrysotile, indicating that

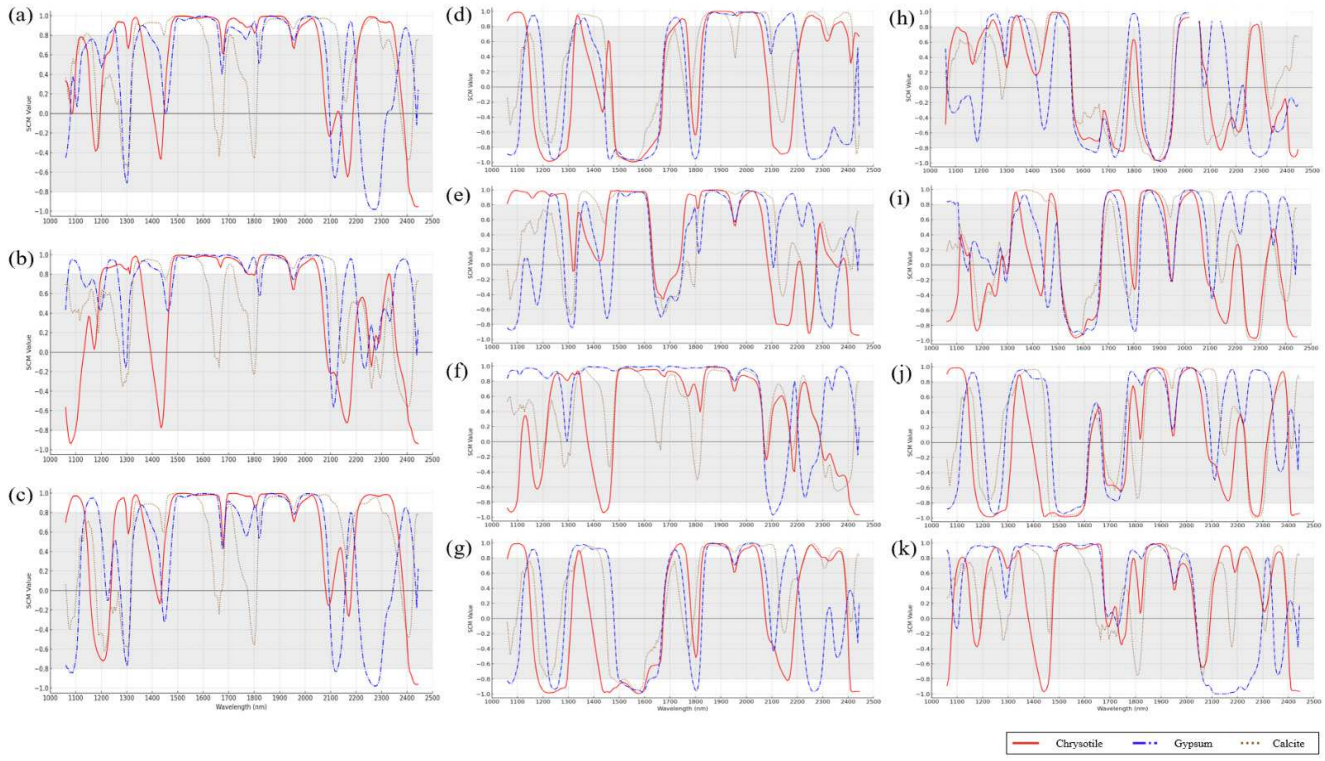


Fig. 13. SCM value distribution showing spectral similarity between indoor ACM and non-ACM and USGS library reference spectra (chrysotile, gypsum, and calcite) across wavelengths. Gray area indicates SCM values from −0.8 to 0.8. (a) ACR, (b) asbestos textile, (c) bamlight, (d) asphalt roof, (e) cement brick, (f) gypsum board, (g) gypsum textile, (h) polystyrene insulation, (i) red brick, (j) tile, and (k) wood panel.

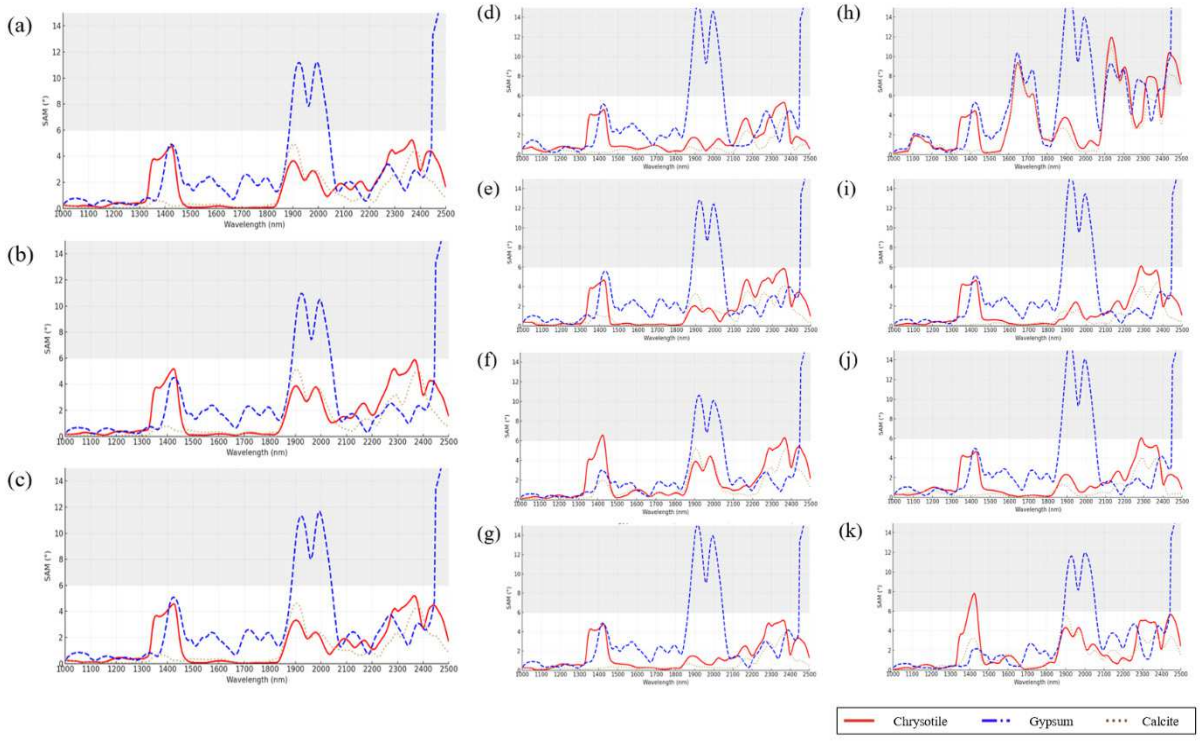


Fig. 14. SAM value distribution showing spectral similarity between indoor ACM and non-ACM and USGS library reference spectra (chrysotile, gypsum, and calcite) across wavelengths. Gray area indicates SAM values greater than or equal to 6°. (a) ACR, (b) asbestos textile, (c) bamlight, (d) asphalt roof, (e) cement brick, (f) gypsum board, (g) gypsum textile, (h) polystyrene insulation, (i) red brick, (j) tile, and (k) wood panel.

mineral-based spectral characteristics were still sufficiently represented. The SAM analysis showed high spectral similarity (below 6°) across all wavelength regions beyond 2000 nm. Around 2350 nm, similar to the SCM results, a high

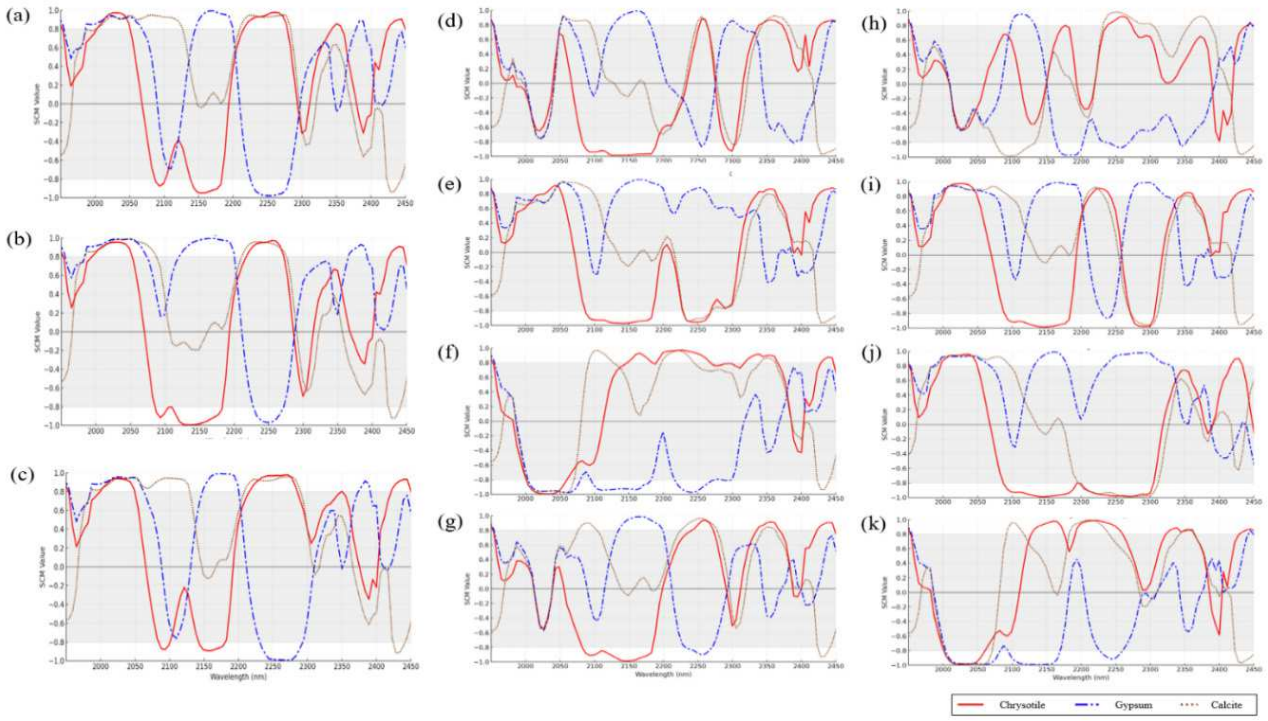


Fig. 15. SCM value distribution showing spectral similarity between outdoor ACM and non-ACM and USGS library reference spectra (chrysotile, gypsum, and calcite) across wavelengths. Gray area indicates SCM values from −0.8 to 0.8. (a) ACR, (b) asbestos textile, (c) bamlight, (d) asphalt roof, (e) cement brick, (f) gypsum board, (g) gypsum textile, (h) polystyrene insulation, (i) red brick, (j) tile, and (k) wood panel.

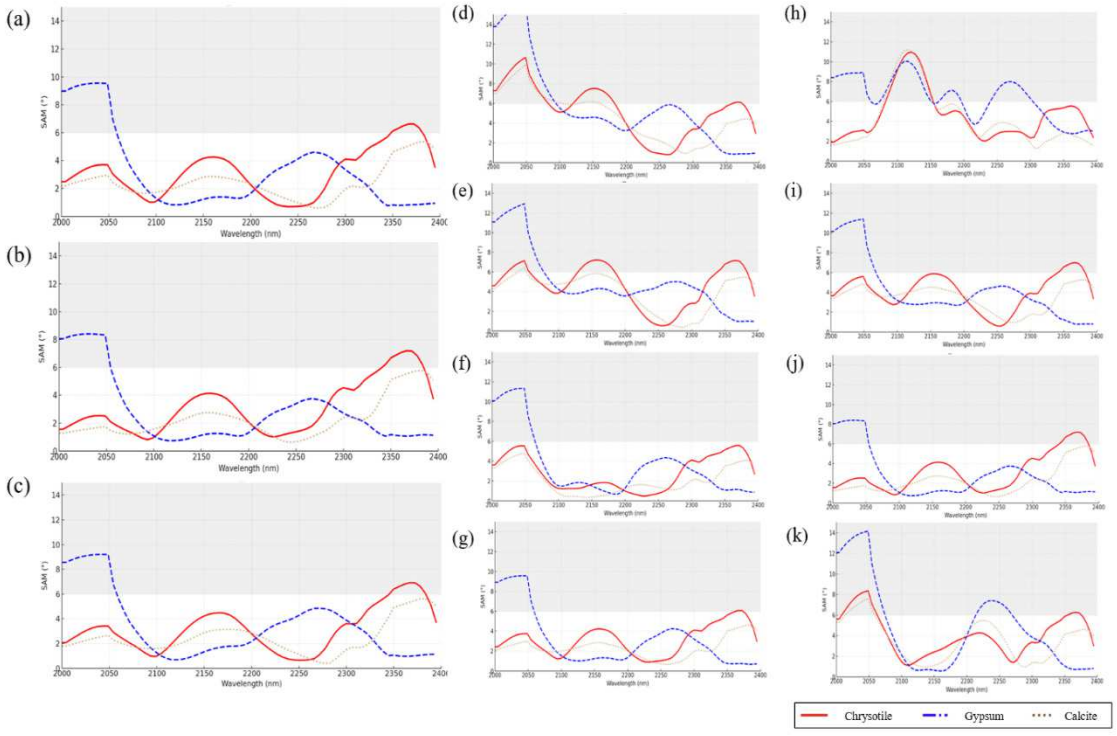


Fig. 16. SAM value distribution showing spectral similarity between outdoor ACM and non-ACM and USGS library reference spectra (chrysotile, gypsum, and calcite) across wavelengths. Gray area indicates SAM values greater than or equal to 6°. (a) ACR, (b) asbestos textile, (c) bamlight, (d) asphalt roof, (e) cement brick, (f) gypsum board, (g) gypsum textile, (h) polystyrene insulation, (i) red brick, (j) tile, and (k) wood panel.

similarity with gypsum was also observed [see Fig. 16(a)–(e)]. Among the non-ACM materials, gypsum board and gypsum textile exhibited notable spectral similarity with chrysotile near 2350 nm according to SCM analysis. Meanwhile, SAM

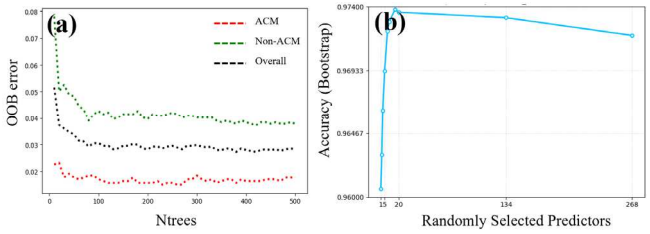


Fig. 17. Variation of (a) OOB error of N_{trees} and (b) accuracy of M_{try} for indoor RF model.

TABLE V

CONFUSION MATRIX OF THE VALIDATION DATA FOR THE CLASSIFICATION RESULTS FROM THE RF CLASSIFICATION MODEL IN INDOOR

	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)
ACM	93.0	98.1	97.0	95.5
non-ACM	97.0	93.0	97.0	95.0
Overall Accuracy : 95.3 % Kappa Coefficient : 0.91				

analysis indicated that all non-ACMs displayed some degree of spectral similarity to chrysotile's absorption features (see Figs. 15 and 16). Throughout the mineral absorption regions, all non-ACM samples demonstrated high spectral similarity with both gypsum and calcite reference spectra. The spectral similarity of indoor and outdoor spectra showed different results for the same material. These results may be caused by spectral complications from atmospheric effects under outdoor conditions and longer acquisition distances. Consequently, materials may display spectral similarity even across different material types. In particular, overlapping absorption regions—such as those related to Mg–OH and CO_3^{2-}—can lead to spectrally similar features among physically distinct materials. Such spectral overlap can negatively impact the classification accuracy of pure spectrum-based algorithms such as spectral feature fitting (SFF) and SAM. Therefore, in spectrally complex environments, more sophisticated algorithms—such as machine learning, deep learning, or probabilistic models that integrate multiple parameters—are essential for extracting reliable detection signals.

C. Classification of CMs Using RF

1) Indoor: The RF classification model for indoor detection was configured with $N_{\text{trees}} = 280$ and $M_{\text{try}} = 268$ based on the OOB error method. The classification results showed a high accuracy of 95.3% with a kappa coefficient of 0.91 for validation data (see Figs. 17 and 18 and Table V). A specificity of 97% shows that the false-positive rate for non-ACM detection is very low. The high $F1$-scores of 95.5% (ACM) and 95.0% (non-ACM) indicate that the model achieves both high precision and recall, allowing it to detect accurately and reliably with few misses and few false detections (see Table V). The important spectral bands for indoor ACM detection derived by Gini impurity overlaps with the absorption features of 1923 ~ 1967 nm, followed by 2299 ~ 2350, 2366 ~ 2426,

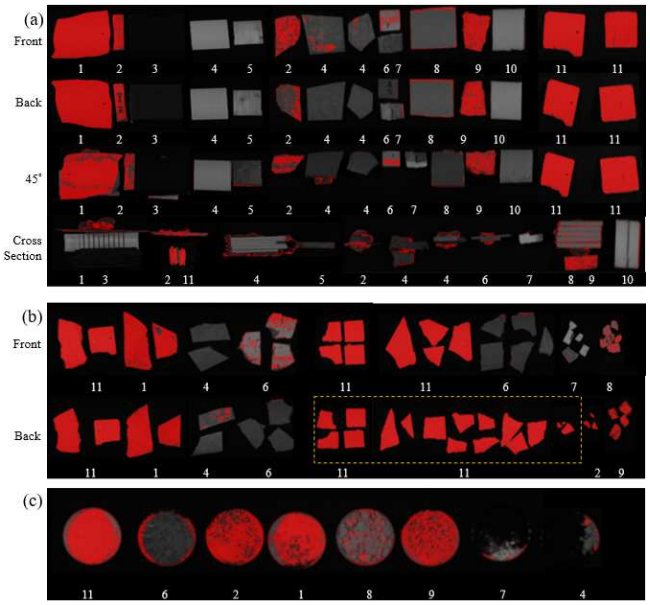


Fig. 18. Mapping results of indoor ACM detection model utilizing RF (a) intact, (b) fragmented, and (c) grinded and powdered. (1) Bamlight, (2) ACR, (3) asphalt roof, (4) tile, (5) wood panel, (6) gypsum textile, (7) gypsum board, (8) red brick, (9) cement brick, (10) polystyrene insulation, and (11) asbestos textile.

TABLE VI

CONFUSION MATRIX OF THE VALIDATION DATA FOR THE CLASSIFICATION RESULTS FROM THE RF CLASSIFICATION MODEL IN OUTDOOR

	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)
ACM	91.5	89.2	99.6	93.5
non-ACM	97.6	98.2	89.3	98.6
Overall Accuracy : 96.5 % Kappa Coefficient : 0.88				

and 2426 ~ 2515, and 1673 ~ 1696 nm (see Fig. 9), corresponding to OH, Mg–OH, CO_3^{2-}, and Fe^{2+}, respectively. The Mg–OH absorption is a unique absorption feature of ACMs manifested by chrysotile, the asbestos mineral. On the other hand, OH absorption is more obvious for non-ACMs, which distinguishes them from ACMs. The CO_3^{2-} band manifested by calcite is a characteristic of specific types, such as cement and gypsum board, of non-ACMs. The Fe^{2+} band showed doublet absorption characteristics in non-ACM polystyrene insulation, but no absorption characteristics were evident in ACMs. The mapping results indicated that ACM detection was generally well performed, with a good detection even in weathered forms (see Fig. 18). The observed misclassifications were mainly FPs in cement bricks and tiles [see Fig. 18(8) and (9)], presumed to have occurred due to the similar absorption characteristics by Al–OH and Mg–OH, where ACM absorption overlaps.

2) Outdoor: The RF model for outdoor ACM detection was configured based on the OOB error method with $N_{\text{trees}} = 500$ and $M_{\text{try}} = 90$. The model for outdoor ACM detection was constructed targeting the spectral region excluding 2000 nm to

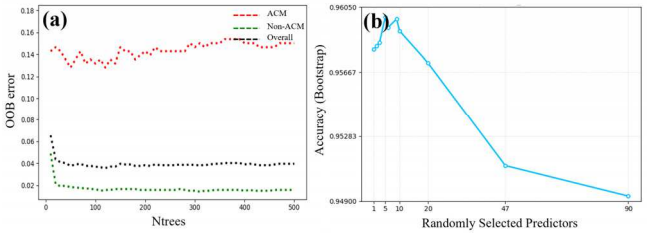


Fig. 19. Variation of (a) OOB error of N_{trees} and (b) accuracy of M_{try} for outdoor RF model.

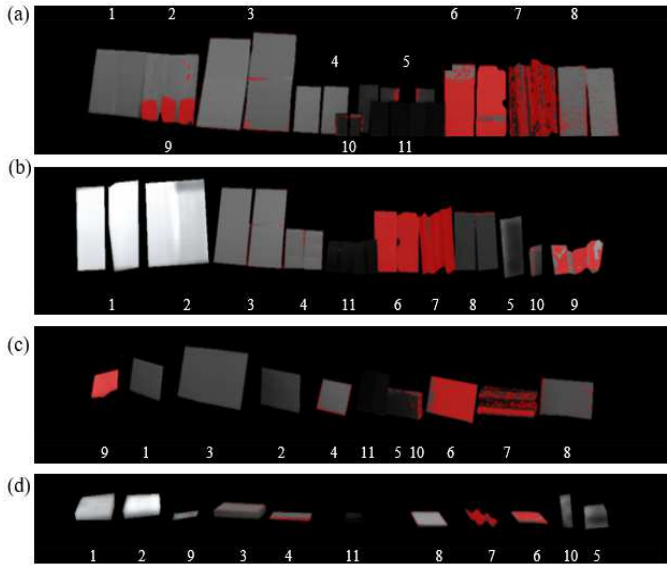


Fig. 20. Mapping results of the outdoor ACM detection model utilizing RF. (a) Front. (b) Back. (c) 45°. (d) Cross section. (1) Polystyrene insulation, (2) wood panel, (3) gypsum board, (4) tile, (5) cement brick, (6) asbestos textile, (7) ACR, (8) gypsum textile, (9) bamlight, (10) red brick, and (11) asphalt roof.

avoid water vapor disturbance. The result shows high detection rates for both ACM and non-ACM, with an OA and kappa coefficient of 96.5% and 0.88, respectively (see Figs. 19 and 20 and Table VI). Specificity values of 99.6% for ACM and 89.3% for non-ACM, along with high $F1$-scores of 93.5% and 98.6%, indicate stable detection performance for both classes. The useful bands for outdoor ACM detection overlap with the absorption features of 2327–2389, 2210–2300, 2204–2288, 2126–2171, and 2003–2015 nm (see Fig. 11), which are associated with Mg–OH, CO_3^{2-}, and Al–OH. The Mg–OH and CO_3^{2-} absorptions also played important roles in indoor ACM detection. Unlike the indoor model, the outdoor model uses Al–OH absorption as one of the most important variables. The Al–OH absorptions are associated with clay minerals found in some of the non-ACMs.

D. Model Construction Using Logistic Regression

1) Indoors: The indoor ACM detection model using BLR was developed based on OH, Fe–OH, Al–OH, Mg–OH, or CO_3^{2-} spectral bands (see Table VII). This selection aligns with spectroscopic analysis results and important variables selected by RF. The constructed model has passed both the

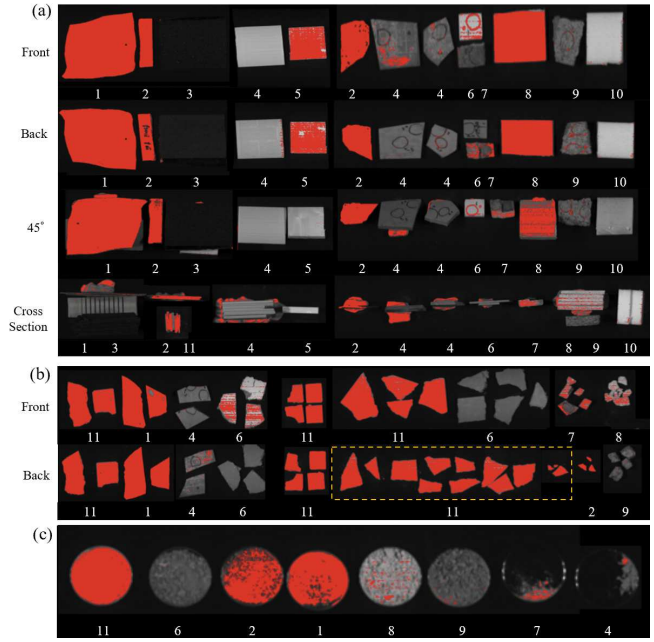


Fig. 21. Mapping results of indoor ACM detection model utilizing logistic regression (a) intact, (b) fragmented, and (c) grinded and powdered. (1) Bamlight, (2) ACR, (3) asphalt roof, (4) tile, (5) wood panel, (6) gypsum textile, (7) red brick, (8) gypsum board, (9) cement brick, (10) polystyrene insulation, and (11) asbestos textile.

TABLE VII
SELECTED WAVELENGTH FOR ACM DETECTION MODEL IN AN INDOOR ENVIRONMENT DERIVED FROM BLR

Wavelength (nm)	β [1]	S.E. [2]	Wald [3]	df [4]	Sig. [5]	Exp(β) [6]
R1546	-21.62	0.00	108.74	1	0.00	1.00
R2205	19.95	0.00	16.14	1	0.00	1.00
R2255	49.39	0.00	60.29	1	0.00	1.01
R2299	78.11	0.00	57.16	1	0.00	1.01
R2316	-123.30	0.00	101.42	1	0.00	0.99
R2349	29.84	0.00	9.31	1	0.00	1.00
R2366	131.37	0.00	29.62	1	0.00	1.01
R2371	-147.43	0.00	46.82	1	0.00	0.99
R2460	6.94	0.00	3.07	1	0.00	1.00
R2493	-20.98	0.00	36.57	1	0.00	1.00
Constant	-2.67	0.00	206.27	1	0.00	0.07

β [1] = logistic coefficient; S.E. [2] = standard error of estimate; Wald [3] = Wald chi-square values; df [4] = degree of freedom; Sig. [5] = p-value; Exp(β) [6] = exponentiated coefficient

TABLE VIII
SUMMARY OF THE BINARY LOGISTIC MODEL FOR INDOOR ACM DETECTION

	Pseudo-R^2		Hosmer and Lemeshow Test		
Parameter	Cox and Snell R^2	Nagelkerke R^2		df	p-value
ACM	0.746	0.995	0.298	8	1

pseudo-R^2 goodness of fit test and the Hosmer and Lemeshow test, indicating its validity (see Table VIII). Furthermore, the validation showed an OA of 86.0% and a kappa coefficient of 0.72 (see Table IX), indicating the model was not overfit to the training data. A specificity of 88.3% and an $F1$-score of 85.6% indicate relatively stable performance in ACM detection. The result obtained by applying the constructed model to the hyperspectral images is shown in Fig. 21. The logistic regression model for indoor ACM detection is defined

TABLE IX
CONFUSION MATRIX OF THE VALIDATION DATA FOR THE CLASSIFICATION RESULTS FROM THE LOGISTIC REGRESSION ACM DETECTION MODEL FOR INDOOR CONDITIONS

	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)
ACM	86.4	86.0	88.3	85.6
Overall Accuracy : 86.0 % Kappa Coefficient : 0.72				

as follows:

$$P_{\text{ACMI}} = \frac{e^{SI_I}}{1 + e^{SI_I}} \tag{11}$$

with

$$\begin{aligned} \text{SI}_I = &-21.62^*R_{1546} + 19.95^*R_{2205} + 043.39^*R_{2255} \\ &+ 78.11^*R_{2299} - 123.30^*R_{2316} + 29.84^*R_{2349} \\ &+ 131.37^*R_{2366} - 147.43^*R_{2371} + 6.94^*R_{2460} \\ &- 20.98^*R_{2493} - 2.67 \end{aligned} \tag{12}$$

where P_{ACM_I} represents the probability of a material being an ACM in indoor conditions, SI_I is the spectral index computed from selected spectral bands and their coefficients, and R_λ denotes the reflectance value at wavelength λ (nm). The model incorporates reflectance values at specific wavelengths (1546, 2205, 2255, 2299, 2316, 2349, 2366, 2371, 2460, and 2493) nm with their corresponding coefficients determined through the logistic regression analysis (see Table VII).

The image results showed that the ACM detection model successfully detected the samples' front, back, cross section, and 45° angles. This indicates that the model can effectively detect ACM from any direction, highlighting its excellent applicability. In addition, the model was able to detect not only intact forms of ACMs but also grinded, fragmented, and powered samples. It also infers that the model can be useful for ACM detections in old buildings and construction trash. Commission error occurred on small portions of non-ACMs, such as tiles, cement bricks, gypsum boards, and wood panels, likely because these materials exhibit absorption characteristics similar to those of chrysotile.

2) Outdoors: Using wavelengths longer than 2000 nm, the model selected ten variables: 2045, 2078, 2083, 2122, 2128, 2177, 2338, 2421, 2476, and 2520 nm (see Table X). Compared with the indoor detection model, the CO_3^{2-} and Mg–OH regions were also used for the outdoor detection model. Unlike the indoor model, the outdoor model selected Al–OH bands, which also showed similar results to the RF model. The model constructed using these variables passed both the pseudo-R^2 goodness of fit test and the Hosmer and Lemeshow test, indicating the validity of the model (see Table XI). The ACM detection model was constructed with an accuracy of 84.3% and the kappa coefficient of 0.68 (see Table XII). Similarly, with a specificity of 87.0% and an $F1$-score of 83.9%, the model demonstrated high classification performance and stable ACM detection (see Table XII).

TABLE X
SELECTED WAVELENGTH FOR ACM DETECTION MODEL IN OUTDOORS DERIVED FROM BLR

Wavelength (nm)	β [1]	S.E. [2]	Wald [3]	df [4]	Sig. [5]	Exp(β) [6]
R2045	-127.33	0.00	197.52	1	0.00	0.98
R2078	28.35	0.00	3.62	1	0.00	1.00
R2083	229.94	0.00	126.53	1	0.00	1.02
R2122	-273.43	0.00	46.86	1	0.00	0.97
R2128	88.42	0.00	4.49	1	0.00	1.00
R2177	74.0	0.00	62.92	1	0.00	1.00
R2338	-18.32	0.00	96.89	1	0.00	1.00
R2421	-6.24	0.00	15.62	1	0.00	1.00
R2476	8.59	0.00	78.06	1	0.00	1.00
R2520	-42.52	0.00	5.45	1	0.00	1.00
Constant	-3.06	0.00	89.46	1	0.00	0.05

β [1] = logistic coefficient; S.E. [2] = standard error of estimate; Wald [3] = Wald chi-square values; df [4] = degree of freedom; Sig. [5] = p-value; Exp(β) [6] = exponentiated coefficient

TABLE XI
SUMMARY OF THE BINARY LOGISTIC MODEL FOR ACM DETECTION IN OUTDOOR CONDITIONS

	Pseudo-R^2		Hosmer and Lemeshow Test		
Parameter	Cox and Snell R^2	Nagelkerke R^2		df	p-value
ACM	0.495	0.659	57.091	8	1

TABLE XII
CONFUSION MATRIX OF THE VALIDATION DATA FOR THE CLASSIFICATION RESULTS FROM THE LOGISTIC REGRESSION MODEL

	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)
ACM	84.4	84.3	87.0	83.9
Overall Accuracy : 84.3 % Kappa Coefficient : 0.68				

The logistic regression model for detecting outdoor ACM is defined as follows:

$$P_{\text{ACMo}} = \frac{e^{SI_o}}{1 + e^{SI_o}} \tag{13}$$

with

$$\begin{aligned} \text{SI}_o = &-127.33^*R_{2045} + 28.35^*R_{2078} + 229.94^*R_{2083} \\ &- 273.43^*R_{2122} + 88.42^*R_{2128} + 74.0^*R_{2177} \\ &- 18.32^*R_{2338} - 6.24^*R_{2421} + 8.59^*R_{2476} \\ &- 42.52^*R_{2520} - 3.06. \end{aligned} \tag{14}$$

Here, P_{ACM_o} represents the probability of a material being an ACM in outdoor conditions, SI_o is the spectral index computed from selected spectral bands and their coefficients, and R_λ denotes the reflectance value at wavelength λ (nm). The model uses reflectance values at specific wavelengths (2045, 2078, 208, 2122, 2128, 2177, 2338, 2421, 2476, and 2520) nm along with their corresponding coefficients, which are determined through the logistic regression analysis (see Table X).

ACM was well detected in the outdoor environment from various directions, like the indoor model; some minor errors occurred with bricks, tiles, and gypsum boards (see Fig. 22). This seems due to the absorption characteristics of calcite

TABLE XIII

CONFUSION MATRICES AND COMPUTATIONAL EFFICIENCY ANALYSIS FOR RF MODEL PERFORMANCE ON REAL-WORLD VALIDATION SITES

	Asbestos textile (1153×384pix, 0.27GB)					Bamlight (989×384pix, 0.22GB)					ACR (492×384pix, 0.20GB)					Town site (7500×384pix, 3.12GB)				
	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)	Processing time	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)	Processing time	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)	Processing time	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)	Processing time
ACM	99.8	85.8	83.6	92.3	2.37 s/img	94.6	90.7	90.4	92.7	3.32 s/img	86.2	100	100	92.6	1.18 s/img	97.7	85.2	98.0	90.9	16.04 s/img
non-ACM	83.6	99.7	99.8	91.0	5363.37 ms/Mpix	90.4	94.4	94.6	92.3	8745.77 ms/Mpix	100	87.8	86.2	83.6	6246.16 ms/Mpix	86.8	98.0	85.2	92.0	5646.59 ms/Mpix
	Overall Accuracy : 91.7 % Kappa Coefficient : 0.83					Overall Accuracy : 92.5 % Kappa Coefficient : 0.85					Overall Accuracy : 93.1% Kappa Coefficient : 0.86					Overall Accuracy : 91.9% Kappa Coefficient : 0.83				

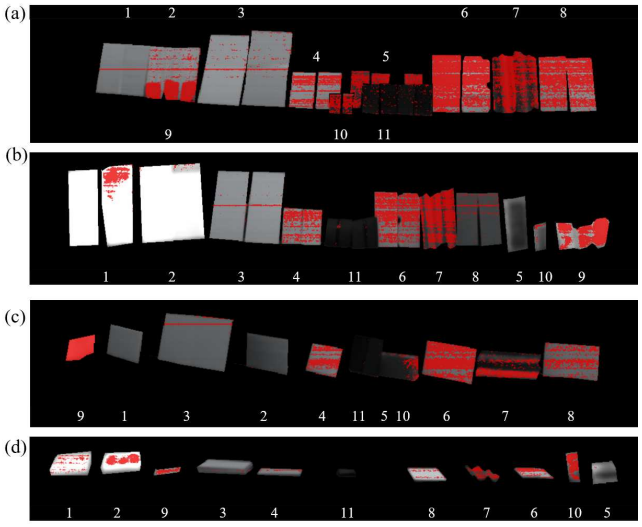


Fig. 22. Mapping results of the outdoor ACM detection model utilizing logistic regression. (a) Front. (b) Back. (c) 45°. (d) Cross section. (1) Polystyrene insulation, (2) wood panel, (3) gypsum board, (4) tile, (5) cement brick, (6) asbestos textile, (7) ACR, (8) gypsum textile, (9) bamlight, (10) red brick, and (11) asphalt roof.

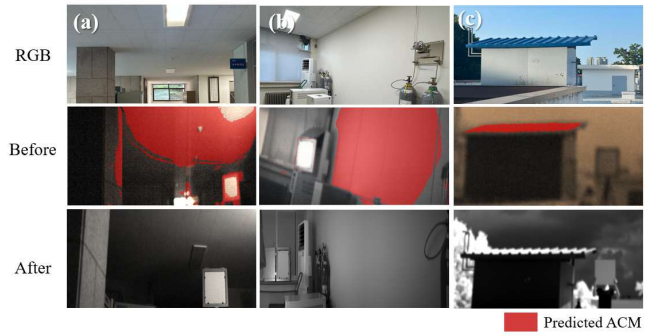


Fig. 23. ACM mapping results from applying the RF ACM detection model to hyperspectral imagery from actual asbestos removal sites showing (a) ceiling, (b) wall, and (c) roof areas before and after removal for feasibility analysis.



Fig. 24. ACM mapping results from the RF detection Model Applied to Hyperspectral Imagery of a town sector in Geumsan City, South Korea.

bands present in bricks, tiles, and gypsum boards, which are similar to the asbestos detection bands.

E. Feasibility Analysis for Field Applicability

The RF-based model, which demonstrated superior performance under both indoor and outdoor conditions, was applied to hyperspectral images from four validation datasets: three sites captured at identical locations before and after asbestos removal, and one large-scale town dataset. To further verify the model's computational efficiency, processing times were comparatively analyzed across all four validation sites.

For indoor conditions, the model achieved an OA of 91.7% with a kappa coefficient of 0.83 for ceilings (asbestos textiles and gypsum textiles), and 92.5% accuracy with a kappa coefficient of 0.85 for interior walls (bamlight and gypsum board). For outdoor ACR and corrugated iron roofing, the model achieved very high classification accuracy with 93.1% OA and a kappa coefficient of 0.86. Furthermore, in the large-scale validation test covering a town sector exceeding 100 m in extent, the model achieved 91.9% OA (kappa: 0.83), demonstrating robust performance even with high-volume datasets (see Table XIII).

The ACM detection model successfully identified most asbestos textile and bamlight areas under preabatement conditions. Although some non-ACM areas were misclassified as ACMs, resulting in relatively higher commission errors, the model demonstrated clear effectiveness in ACM detection [see Fig. 23(a)]. Under postabatement conditions, no pixels were detected as ACMs, confirming successful removal verification [see Fig. 23(b)]. Notably, structural elements such as pillars and lighting fixtures within the scene were not replaced during abatement; yet, some pixels associated with these features were classified as ACM in preabatement imagery and non-ACM in postabatement imagery. This inconsistency is likely attributed to varying illumination conditions, where side-angle sunlight can introduce spectral noise through bidirectional reflectance distribution function (BRDF) effects [51]. The postabatement imagery was acquired under rainy conditions, while preabatement imagery was captured under sunny conditions with higher solar irradiance entering through windows, resulting in increased spectral variability.

For outdoor conditions, ACR boundaries were clearly delineated, with almost all ACR areas correctly identified as ACM in preabatement imagery, while no ACM pixels were detected in the replaced corrugated iron roofing in postabatement

TABLE XIV
OBJECT-BASED ASSESSMENT OF ACM DETECTION PERFORMANCE ACROSS TEST SITES

Test Sites	Total Objects	Detected Objects	Detection Rate (%)	False Alarm Rate (%)
Indoor	14	2(TP) + 1(FP)	100	8.33
Outdoor	4	1(TP) + 0(FP)	100	0.00
Town site	17	1(TP) + 1(FP)	100	6.25

imagery [see Fig. 23(c)]. This high-accuracy performance demonstrates the method's practical value for abatement project assessment, efficiently verifying removal completeness across large rooftop areas without requiring extensive manual sampling or visual inspection.

In the large-scale imagery, ACM was generally well detected, although some cement-coated roofs and walls were occasionally misclassified as ACM. Of the 17 buildings analyzed, only one contained an ACR, which was successfully detected in its entirety (Fig. 24). Some cement walls and roofs exhibited partial false-positive classifications. The high user's accuracy confirms the model's reliability for ACM identification, while the slightly lower producer's accuracy reflects occasional misclassification of cement-based materials. Critically, true ACM surfaces were completely detected, whereas FPs appeared only as isolated patches easily distinguished through visual verification.

Processing performance evaluation showed that the smallest dataset (0.22 GB) required approximately 1.18 s for detection at 6246.16 ms/MPixel, while the large-scale dataset (3.12 GB) required 16.04 s at 5646.59 ms/MPixel (see Table XIII). The consistent per-megapixels processing rates across different data volumes demonstrate that the model maintains high computational efficiency at scale, making it well-suited for near real-time ACM detection applications.

The object-based detection assessment demonstrated superior performance compared to pixel-based metrics across all test cases. As shown in Table XIV, the method achieved a 100% detection rate, successfully identifying all ACM objects. Furthermore, the false alarm rate remained remarkably low at approximately 6%, indicating minimal misclassification of non-ACM structures as ACMs. These object-level results exceed the pixel-based performance metrics and significantly enhance the method's practical applicability for real-world scenarios, where ACMs typically exist as complete structural elements rather than isolated pixels. This robust object detection capability ensures reliable identification of ACMs in building components such as walls, roofs, and ceilings, providing critical support for regulatory compliance and remediation efforts.

This approach provides unique capabilities for ACM detection spanning individual buildings to large-scale surveys. Our method improves upon conventional visual inspection and sampling-based approaches by delivering a nondestructive, spatially continuous assessment that simultaneously maps ACM distribution and verifies removal completeness. The object-based assessment framework complements pixel-level analysis, offering additional perspective on detection performance in practical scenarios, where ACMs typically exist as structural elements. The validated before-and-after comparison framework offers an objective tool for evaluating abatement effectiveness and ensuring regulatory compliance. The consistent computational efficiency demonstrated across varying data volumes—from single rooms to entire town sectors—establishes the method's viability for large-area surveys in diverse urban and rural settings with near real-time processing. This approach effectively bridges the gap between spectral classification accuracy and practical field applications, supporting reliable identification of ACMs in real-world environments.

IV. DISCUSSION

This research represents the most comprehensive study to date on ACM detection using hyperspectral imaging, with several key innovations and contributions to the field. Our work is the first to develop detection models that consider multiple real-world scenarios, including various material forms, viewing angles, and environmental conditions. Through extensive validation using XRD, SEM, and EDS analyses, we confirmed that chrysotile is the predominant asbestos mineral in ACMs and demonstrated that the combination of Mg and Si elements serves as a reliable indicator for asbestos presence in CMs.

A significant innovation of our study is the development and validation of specialized models for both indoor and outdoor environments with demonstrated real-world applicability. The indoor detection model achieved remarkable accuracy using the RF algorithm (95.3%), particularly by identifying key absorption features at 2299–2350, 1923–1967, and 2366–2426 nm, corresponding to Mg–OH, OH, and CO_3^{2-} absorptions. This advancement enables more reliable detection across various material conditions, including weathered, powdered, and fragmented ACMs. Compared with the previous work, our indoor model did not select the 1380–1400-nm absorption feature by OH [26], which had been commonly used in earlier studies, indicating improved feature selection tailored to our sample diversity.

For outdoor detection, our RF model achieved 96.5% accuracy by incorporating additional Fe–OH and Al–OH absorption bands alongside the Mg–OH and CO_3^{2-} bands used in the indoor model. This represents a significant improvement over previous outdoor studies that primarily focused on roof-only material detection using only the Mg–OH band. Our comprehensive approach, using a larger and more diverse dataset than previous works, has resulted in more robust and versatile detection capabilities. This represents a significant improvement over previous outdoor studies, which primarily focused on roof-only material detection using only the Mg–OH band. Our model, in contrast, was developed using a more diverse dataset containing a wider range of ACM forms and surface conditions, resulting in a broader spectral basis for classification.

Real-world applicability testing of the model across three validation sites from asbestos abatement projects demonstrated excellent performance, achieving OAs of 92.1% (κ = 0.84) for indoor conditions and 93.1% (κ = 0.86) for outdoor conditions. The model successfully identified most asbestos

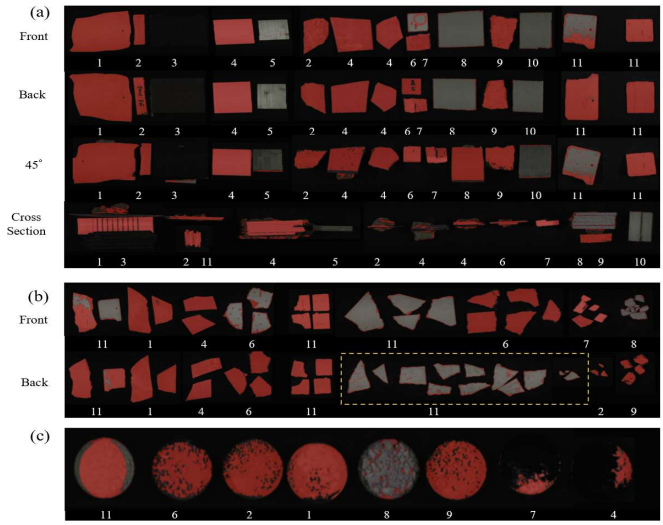


Fig. 25. Comparative ACM detection mapping results obtained using the benchmark PLS-DA model (modified from Bonifazi et al. [26]) under indoor conditions showing (a) intact, (b) fragmented, and (c) ground and powdered samples. Materials: (1) bamlight, (2) ACR, (3) asphalt roof, (4) tile, (5) wood panel, (6) gypsum textile, (7) red brick, (8) gypsum board, (9) cement brick, (10) polystyrene insulation, and (11) asbestos textile.

TABLE XV

CONFUSION MATRIX FOR PLS-DA CLASSIFICATION MODEL (MODIFIED FROM BONIFAZI ET AL. [26]) SHOWING ACM VERSUS NON-ACM DETECTION PERFORMANCE UNDER INDOOR AND OUTDOOR TEST CONDITIONS

Indoor				
	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)
ACM	55.2	83.0	32.6	66.4
non-ACM	65.7	32.6	83.0	43.5
Overall Accuracy : 57.8% Kappa Coefficient : 0.16				
Outdoor				
	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)
ACM	61.8	75.4	53.4	67.9
non-ACM	68.5	53.4	75.4	60.0
Overall Accuracy : 64.4% Kappa Coefficient : 0.29				

textile, bamlight, and ACR areas under preabatement conditions, and crucially, detected no ACM pixels postabatement, confirming successful removal verification. While some commission errors occurred due to misclassification of non-ACM areas as ACMs, the model's clear effectiveness in ACM detection and removal verification demonstrates significant practical value for abatement project assessment. Further validation on a large-scale test site encompassing 17 residential structures within a 100-m sector yielded comparable accuracy (91.9%, κ = 0.83), establishing the model's scalability for comprehensive ACM surveys. Computational performance analysis indicated consistent processing efficiency across all validation sites (5646.59–6246.16 ms/MPixel), confirming the algorithm's suitability for operational deployment in real-world monitoring scenarios.

This approach offers substantial advantages over conventional visual inspection or sampling-based methods by providing nondestructive, spatially comprehensive assessment capabilities that can simultaneously map ACM distribution and verify removal completeness across large areas without extensive manual intervention. The method's applicability across diverse environments and its high-accuracy performance represent a significant advancement in asbestos detection technology, establishing an efficient framework for compliance verification and quality assurance in real-world abatement scenarios. Furthermore, its demonstrated scalability—from single rooms to entire town sectors—with consistent computational efficiency establishes the method's viability for large-area surveys in diverse settings with near real-time processing capability.

To assess the relative effectiveness of our approach, we conducted a comparative analysis implementing the methodological framework of Bonifazi et al. [26], which employs PCA and PLS-DA techniques, using a material classification scheme similar to that used in the previous research. The comparative analysis revealed that our proposed model outperformed the benchmark model, achieving 95.3% versus 57.8% accuracy for indoor conditions and 96.5% versus 64.4% accuracy for outdoor conditions (see Table XV). The benchmark model's inferior performance was primarily attributed to excessive FPs, failing to detect critical ACM features while frequently misclassifying non-ACM materials such as gypsum textile, red brick, cement brick, and tiles as ACMs (see Figs. 25 and 26). When applied to real-world asbestos removal sites, the benchmark model failed to detect ACMs with accuracy below 50%, showing limitations of existing methods when transitioning from controlled laboratory conditions to complex field scenarios (see Fig. 27 and Table XVI). In contrast, our proposed model maintained consistently high accuracy across real-world datasets and demonstrated discrimination capability between preasbestos and postasbestos removal conditions, validating its practical utility and reliability for real-world ACM detection applications.

Despite these advancements, several limitations require consideration. The controlled laboratory and outdoor testing conditions, along with validation at a single real-world site, may not fully represent the diversity of field scenarios. While our models performed well at the tested detection distance of 5.3 m, applications requiring longer range detection may encounter additional challenges. We observed potential misclassification issues under varying illumination conditions, particularly in indoor environments where side-angle sunlight introduces spectral noise through BRDF effects. Future research should focus on expanding practical implementation through UAV platforms capable of operating at extended detection ranges (10–200 m). Establishing a comprehensive database incorporating field-acquired data under various illumination conditions from actual ACM sites would substantially enhance model applicability. Integration of accumulated empirical data with advanced deep learning architectures and unified modeling frameworks could optimize detection accuracy across diverse environmental conditions. This research establishes a solid foundation for practical hyperspectral imaging applications in ACM detection. With continued development addressing these limitations, these methods could

TABLE XVI

CONFUSION MATRIX SHOWING CLASSIFICATION PERFORMANCE OF THE PLS-DA MODEL (MODIFIED FROM BONIFAZI ET AL. [26]) APPLIED TO REAL-WORLD ASBESTOS REMOVAL SITE FOR FEASIBILITY ANALYSIS

	Asbestos textile				Bamlight				ACR			
	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)	User's Accuracy (%)	Producer's Accuracy (%)	Sensitivity (%)	Specificity (%)	User's Accuracy (%)	Producer's Accuracy (%)	Specificity (%)	F1-score (%)
ACM	0.0	0.0	80.0	0.0	0.0	0.0	100	0	0.0	0.0	99.2	0.0
non-ACM	44.7	80.8	0.0	57.7	50.0	100	0	66.7	49.8	99.2	0.0	66.3
	Overall Accuracy : 40.4 % Kappa Coefficient : -0.19				Overall Accuracy : 50 % Kappa Coefficient : 0.0				Overall Accuracy : 49.6% Kappa Coefficient : -0.008			

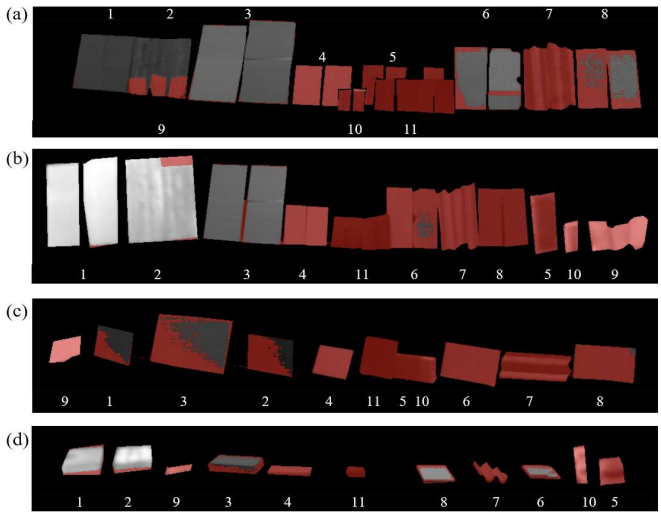


Fig. 26. Comparative ACM detection mapping results obtained using the benchmark PLS-DA model (modified from Bonifazi et al. [26]) under outdoor conditions showing (a) front, (b) back, (c) 45°, and (d) cross section. (1) polystyrene insulation, (2) wood panel, (3) gypsum board, (4) tile, (5) cement brick, (6) asbestos textile, (7) ACR, (8) gypsum textile, (9) bamlight, (10) red brick, and (11) asphalt roof.



Fig. 27. Results of benchmark PLS-DA ACM detection model (modified from Bonifazi et al. [26]) applied to actual asbestos removal sites (a) ceiling, (b) wall, and (c) roof areas before and after removal.

significantly improve public health and safety through more efficient and reliable ACM detection and management in both indoor and outdoor environments.

V. CONCLUSION

This study establishes the most comprehensive and reliable framework for ACM detection using hyperspectral imaging technology. Our detection models were developed using the most diverse and representative sample dataset of real-world cases, encompassing both indoor and outdoor settings with materials observed from multiple viewing angles and prepared in various real-world forms. The indoor model achieved 95.3% OA, and the outdoor model demonstrated 96.5% accuracy.

The indoor model identified critical absorption features at 2299–2350, 1923–1967, and 2366–2426 nm as most effective for distinguishing ACM from non-ACM materials, corresponding to Mg–OH, OH, and CO_3^{2-} absorptions. The outdoor RF model incorporated additional Fe–OH and Al–OH bands for enhanced discrimination capability. Through comprehensive validation using XRD, SEM, and EDS analyses, we confirmed that chrysotile is the predominant asbestos mineral in ACMs and established that the combination of Mg and Si elements serves as a reliable indicator for asbestos presence in CMs.

Practical validation at three abatement sites and one large-scale town site achieved OAs ranging from 91.7% to 93.1%. The model successfully identified ACMs and verified complete removal postabatement. Notably, it accurately detected the single ACR roof among 17 buildings in the town sector, confirming scalability with high-volume data. Consistent computational efficiency across data volumes demonstrates practical viability for real-world applications. Comparative analysis with existing benchmark methodologies confirmed our approach's superior performance, achieving 95.3% versus 57.8% accuracy for indoor conditions and 96.5% versus 64.4% accuracy for outdoor conditions. While benchmark models exhibited excessive FPs, our proposed models maintained consistently high accuracy across real-world datasets and demonstrated reliable discrimination between preabatement and postabatement conditions.

Our dual-level assessment framework represents a significant advancement in asbestos management technology, offering nondestructive, spatially comprehensive detection that enhances conventional visual inspection methods. By evaluating performance at both pixel and object levels, we provide a more complete understanding of detection capabilities in practical scenarios where ACMs typically exist as structural elements. This approach delivers immediate practical value for compliance verification and quality assurance in real-world abatement scenarios and large-scale surveys, providing objective documentation for regulatory compliance while substantially reducing costs compared with conventional destructive sampling methods. Our framework demonstrates consistent computational efficiency across varying spatial

scales, from individual buildings to entire town sectors. The framework's reliable performance across diverse test sites establishes its potential for future UAV platform integration, enabling extended-range detection applications. This research establishes a robust foundation for enhanced public health and safety through more efficient, cost-effective, and reliable ACM detection and management across diverse environmental conditions.

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Hyeji Sim received the B.S. degree in geology and Earth environmental sciences and the M.S. degree from the Department of Astronomy, Space Science and Geology, Chungnam National University, Daejeon, South Korea, in 2023 and 2025, respectively.

Her research interests focus on mineral spectroscopy and hyperspectral remote sensing for environmental material detection, particularly asbestos identification in building materials.



Jaehyung Yu received the B.S. degree in geology and the M.S. degree in applied geology from Chungnam National University, Daejeon, South Korea, in 1997 and 1999, respectively, and the Ph.D. degree in geography from Texas A&M University, College Station, TX, USA, in 2005.

He was an Assistant Professor and an Associate Professor with Texas A&M University–Kingsville, TX, USA, from 2005 to 2012. He moved to the Department of Geology and Earth Environmental Sciences, Chungnam National University, in 2012, where he is currently a Professor. His research interests include uncrewed aerial vehicle (UAV) and satellite remote sensing applications on geological processes.



Lei Wang received the Ph.D. degree in geography from Texas A&M University, College Station, TX, USA, in 2006.

He is currently an Associate Professor with the Department of Geography and Anthropology, Louisiana State University, Baton Rouge, LA, USA. His research interests include remote sensing applications, image analysis, geostatistical models, surface hydrology, and flood hazards.



Chanhyeok Park received the M.S. degree in geology and Earth environmental sciences from Chungnam National University, Daejeon, South Korea, in 2020, where he is currently pursuing the Ph.D. degree with the Department of Astronomy, Space Science and Geology.

His research interests include uncrewed aerial vehicle (UAV) remote sensing, machine learning, especially multi-image and hyperspectral image processing.



Huy Hoa Huynh received the B.S. degree in geology and earth environmental sciences and the M.S. degree in applied geology from Chungnam National University, Daejeon, South Korea, in 2020 and 2022, respectively, where he is currently pursuing the Ph.D. degree with the Department of Astronomy, Space Science and Geology.

His research interests include remote sensing applications, machine learning, especially ground-based hyperspectral image processing, and mineral resources exploration.



Hyun-Cheol Kim (Member, IEEE) received the Ph.D. degree in Earth and environmental sciences (with a major in satellite oceanography) from Seoul National University, Seoul, South Korea, in 2006.

Since 2007, he has been with the Korea Polar Research Institute (KOPRI), Incheon, South Korea, using satellite and uncrewed aerial vehicle remote sensing data for the purpose of Arctic and Antarctic research. He established a Remote Sensing Group, KOPRI, in 2016. He leads the Center of Remote Sensing and GIS as the Director.